
# Final Report

Erebus Mission Team 2

ASEN 5158 – Space Habitat Design

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# Introduction and Overview

For our project, we will be creating and defining the parameters for a Low Earth Orbit (LEO) mission housing up to three crew members at a time, to conduct ground-breaking research on radiation. Our team will be known as the “Erebus Mission”, mimicking the space shuttle’s naming scheme of notable exploration ships. HMS *Erebus* was a British exploration vessel that contributed to the early exploration efforts of the Antarctic and Arctic, embodying how our space station will be multi-polar. Erebus is also the name of the Greek primordial god of darkness and shadows, representative of the elusiveness of radiation. Our station will be named “Boyd Station” after Louise Boyd, who was an American explorer. She was the first woman to fly over the North Pole and documented her journeys extensively. Our mission will be dedicated to her for her being a pioneering woman in aviation, and her studies of the North Pole, a scarcely characterized region at the time.

One of the primary goals of our mission will be implementing a sun-synchronous polar orbit. This will allow for unique data collection, due to the distinctive radiation environment/climate observation potential. Access to Earth’s polar magnetosphere, boreal surface climates, and polar ranges of radiation belts are all scientifically significant environments. All of these environments would benefit from human interaction in orbit, whether it be short term radiological studies, experimenting with Earth observation hardware, or long-term configurable space weather experiments. We believe this will give the mission the necessary edge to convince our stakeholders of its importance.

Our team will be broken into eight roles: Project Manager, Mission Engineer, Systems Engineer, ECLSS, EVA/CA/PA, Power/Thermal, Structures, and Avionics. Over the course of three months, we plan to outline - at minimum - the bare bone requirements for our habitable space mission and create estimations for critical supplies/resources. Basic analysis will be conducted to determine structural integrity and orbit longevity to sustain the mission for at least five years. Specifications surrounding high priority items, like the EVA, will be outlined in greater detail.

Collectively, we believe that conducting in-depth radiation research will greatly benefit our society, as concerns increase surrounding its impact on astronauts in space and humans on the ground. Passing through the polar regions of the radiation belts allows for crew and other biological subjects to be exposed to predictable and short periods of radiation to study the effects of exposure. Therefore, the station itself can be used as a testbed to prove out radiation countermeasure methodologies, which could be more permanently applied to longer duration deep space missions.

## Mission Goals

The mission aims to support three crew members in a reusable space habitat during ten recurring seven-day missions. The habitat will be launched uncrewed into a 500 km circular orbit to operate as a commercial LEO station. After successful orbital insertion and system checkouts, the station will be ready to receive the first crew and commence nominal crewed mission operations.

Sometime between day four and six of the mission, two of the three crew members will conduct a three-hour extravehicular activity (EVA). This EVA will be umbilical based, meaning the astronauts will remain physically connected to the station through a tether system that provides power, communications, and life support. Specific EVA tasks will be defined by the visiting crew on a per-mission basis. At the conclusion of the seven-day mission, the crew will transfer aboard the docked transport vehicle and depart from the station.

## Ground Rules

The prescribed ground rules for the space habitat mission are listed below:

1. Habitat must be launched into a 500 km circular parking orbit in a 'ready for occupancy' state with initial checkout assumed (i.e., no on-orbit assembly required, but inflatable options can be considered). Any cargo necessary for the first mission should be launched within the habitat.
2. Initial habitat launch must include all provisions/consumables/expendables needed for the seven-day, three-person mission.
3. Transportation of the crew from Earth to the habitat in LEO is provided independently outside the scope of this effort, with return to Earth similarly provided at mission end, also from the 500 km circular orbit.
4. Habitat must enable pressurized docking or berthing with the crew transport vehicle for arrival and departure transfer operations.
5. Habitat must enable one three-hour, umbilical-based EVA for two crewmembers while on orbit.
6. Spacesuits for EVA will be provided by arriving crews.

7. Regenerable life support processes are not permitted (i.e., all metabolic consumables are delivered at the start of each mission).

8. A means of jettisoning all waste products at the end of the mission must be provided. Waste does not need to be safely returned to the surface of the Earth.

9. Support integration and operation of four MLE (middeck locker equivalent) payloads.

10. Must select an existing launch vehicle (or currently planned by 2030) capable of delivering the fully loaded but uncrewed initial habitat from Earth to a 500 km orbit.

11. An onboard propulsion system will provide sufficient delta-v to maintain 500-km orbit for five years. The orbit may not deviate from 500-km by more than +/-25 km.

12. The habitat must be reusable for at least ten missions, starting no later than 2030, and at least five years. Resupply logistics will be provided by the subsequent crew transport vehicles. However, the habitat should remain in a 'ready for occupancy' state. Specifically, when each subsequent mission's crew arrives, the habitat should have a suitable atmosphere, though oxygen or other atmospheric gas resupply to supplement that atmosphere can be provided by subsequent crew transport vehicles. Limited maintenance of the habitat can be provided by subsequent crew transport vehicles.

13. Habitat must be deorbited for destruction via atmospheric entry within one month of final mission. Deorbit capabilities should be part of the habitat and not provided by a crew transport vehicle. That habitat does not need to be safely returned to the surface of the Earth after the final mission.

## Assumptions

1. **Re-entry vehicle can precisely target landing site**

- A polar orbit has periods where it is not necessarily above an adequate landing site, so the reentry capsule will need to be able to safely target an adequate site, perhaps by traversing cross-range.

2. **Crew capsule will remain docked to the station for the duration of the mission**

- A continuously docked capsule can provide emergency radiation and Micrometeoroid and Orbital Debris (MMOD) shelter, and depending on future analysis, potentially provide a pressurized section for the third non-EVA-participating crewmember to remain during the EVA.

3. **Station is exclusively free-flying and single launch**

- Exclusion of the possibility of additional modules to the space station and no need for separate resupply craft means that only a single docking port is required to be designed into the station.

4. **Orbit-keeping propellent is stable through the mission and refueling is not required**

- Assuming a shelf-stable rocket propellent allows for simpler calculations of propellent mass, and eliminating the possibility of refueling simplifies the design and location of fuel tanks. This assumption is flexible based on available data concerning propellant decay rates and if margins can be appropriately accounted for.

5. **Station is not permanently occupied**

- While part of the ground rules involves temporary missions, this assumption allows for ECLSS, power, and radiation shielding to be designed for shorter-term, intermittent use and any active systems to not be maintained while the station is unoccupied.

6. **Atmospheric conditions, solar weather, cosmic ray flux, and MMOD incidence do not deviate significantly from standard**

- Radiation shielding and orbit-keeping maneuver calculations require an assumption or model of radiation incidence and drag. Assuming consistent standard conditions simplifies these calculations with an addition of safety margin for unforeseen events. Space weather models will be based on the predicted solar cycle range experienced during the years of station operation.

7. **Components degrade at a standard rate**

- Consistent solar wind and MMOD collisions allow for simpler design-for-demise calculations, as degradation of performance will be consistent across the lifetime of the spacecraft.

8. **Collision avoidance maneuvers occur and must be budgeted for in propellent load**

- In addition to maintaining a stable orbit, it can be assumed that maneuvers must be made to avoid large orbital debris, and that propellent to make those maneuvers come from the station itself and not from visiting spacecraft.

9. **EVA suit umbilical and linkages are standardized and can interface with station**

- Designing the umbilical linkage is outside the scope of this project and will be assumed to work adequately for the mission.

10. **Human crew is competent and capable of performing all maintenance tasks**

- It is likely that the station will require periodic maintenance, and it is reasonable to assume that the crew will be capable of handling any planned repairs or upkeep deemed necessary, such as replacing consumable ECLSS components or analyzing performance of components.

11. **Visiting capsule will require ECLSS and power support from the station**

- It is common for capsules visiting the ISS to turn off their own management systems and rely on the ISS for power, ECLSS, and thermal controls, so it is reasonable to assume that this station will also be required to provide such support for visiting spacecraft.

## Project Objective

Our team defined both high level and low functional objectives within a Functional Decomposition (FD), with inclusions of information such as mass estimates when necessary. Members within our Mission Engineering group oversaw the creation of our mission Concept of Operations (ConOps) diagram and CA/PA/EVA designed a hypothetical Operational Concept (OpsCon) for our first planned mission aboard our space habitat. Team members within our Mission Engineering group also developed a Design Mission Reference. We designed a Master Equipment List (MEL) that includes mass, power, and volume based on the environmental needs of the three crew members in their seven-day mission as well as customer needs to complete their goals. We also designed a habitable volume and the corresponding pressurized volume based on a Celentano curve-like calculation, needs for structures, and the MEL's estimation non-habitable volume. A launch site and vehicle for the habitat was chosen based on the intended orbit, pressurized volume, and habitat mass. A separate calculation was done to maintain the orbit for the five-year period. The Structures team created volumetric sketches and CAD models based on the requirements of each sub-team. The Structures and Mission Engineering sub-teams considered radiation risks in the chosen orbit and potential shielding methods. Additionally, each sub-team considered the risks within their domain and their level of consequence.

Finally, our team compared our habitat and business model against the Polaris Dawn and similar space station habitat missions such as Skylab, Salyut, and Vast Haven-

1. Based on comparisons between prior missions and our design, we developed a business model concept relevant to our mission goals and team decisions. To maintain a regular schedule for project tasks and deliverables we created and followed a Gantt chart (see Figure 2).

## Org Chart

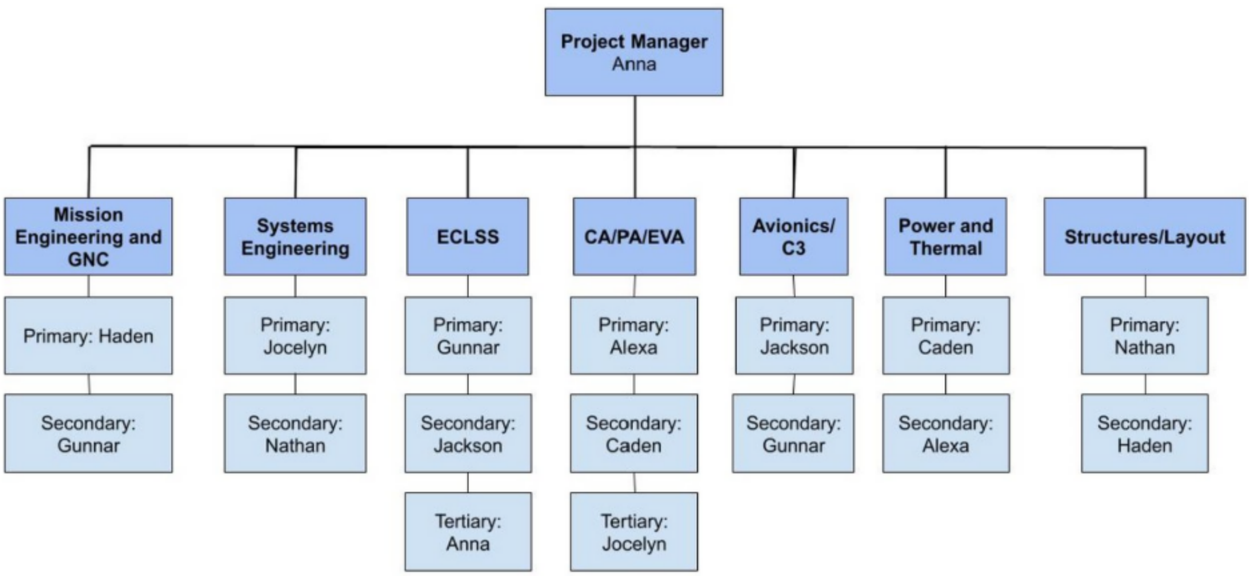


*Figure 1111. Organizational Chart for the Erebus Mission.*

## Gantt Chart

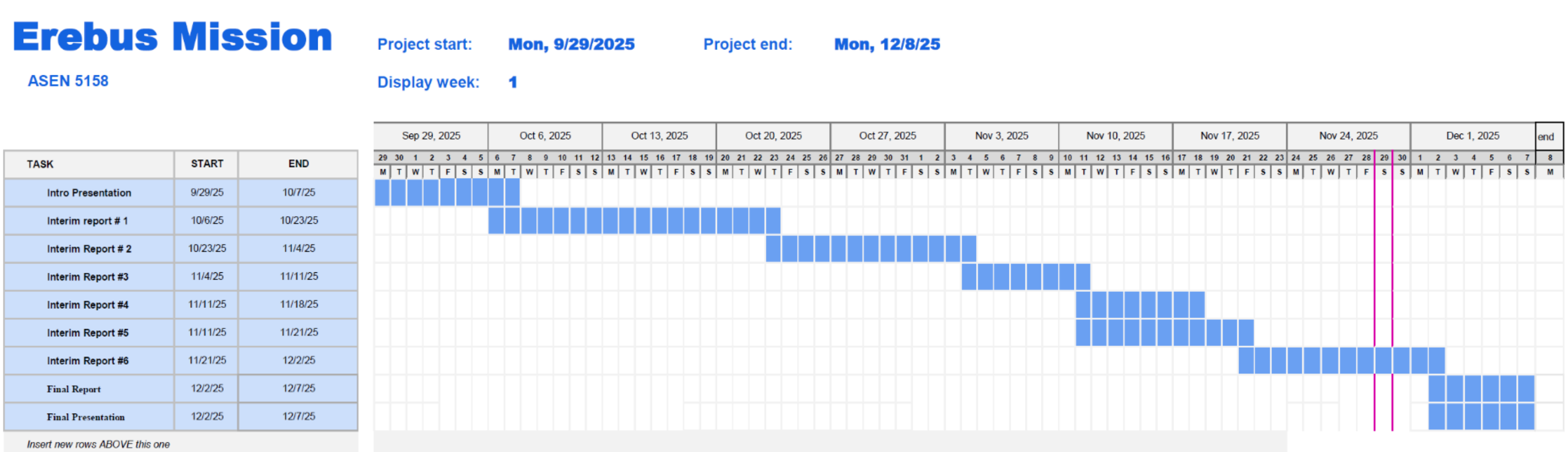


*Figure 2222. Erebus mission Gantt chart.*

# Mission Engineering

## High Level Functional Objectives (FOs)

*Table 1111. High level FOs for Boyd Station.*

| **High Level FOs** |
|---|
| Launch habitat |
| Enable crew ingress |
| Keep the crew alive |
| Perform a two person EVA |
| Maintain polar orbit for five years |
| Incorporate a means for managing waste |
| Design the habitat for demise |
| Allow for integration and operation of four middeck locker payloads |
| Enable constant ground communication |
| Maintain nominal operation of the station during occupation and nonoccupation |
| Maintain Habitat reusability for five years |

## Design Reference Mission (DRM)

### Preflight Operations

#### *Launch Site and Payload Processing*

The station will arrive prior to launch to begin final checkout and integration with the launch vehicle. Final testing of the station during integration process will verify that systems are ready for launch. During this period, commodities will be loaded onto the spacecraft. Following launch vehicle integration, the vehicle will be rolled out to the launch pad and prepared for launch. During this period the station will begin to be monitored for conditions like environmental quality and Command, Control and

Communication (C3) system operation. Vehicle and payload will perform closeout preparations a few days in advance of launch. Launch of Boyd Station can be best facilitated from the Western Test Range (WTR), out of Vandenberg Space Force Base.

### *Crew Training and Launch Preparation*

While the station is performing final checkout and vehicle closeout procedures, the first visiting crew and their spacecraft will be preparing for launch as well. Crew will enter a two-week quarantine prior to launch to ensure their health on-orbit. The launch vehicle will begin its closeout procedures around the time that Boyd Station performs its launch to orbit. This staggering of the launch of station and crew is to ensure ample time to perform station checkout and certify the station is suitable for habitation. Crew launch will occur out of Kennedy Space Center (KSC) as it is the United States' only man rated launch site.

The visiting crew will have already been trained on Boyd Station's various functions and means of operation in conjunction with ground control crew. They will know how to perform a rendezvous and docking procedure with their spacecraft, operate the environmental control systems, perform the scientific experiments on board and prepare the station for a scheduled extravehicular activity (EVA). This training will occur in the years and months leading up to the station's launch, and the crew's input might prove valuable in making the station accommodating for use. Additionally, ground crew will be trained on the operation of the station to support the crew through their time on-orbit.

## Station and Primary Crew Launch

### *Station Launch and Checkout*

Boyd Station will lift off aboard a carrier vehicle to be placed in a 500 km Sun Synchronous Orbit (SSO). Following burnout and insertion of the station into orbit, the station will perform a multi-day checkout procedure. The orbit of the station will be verified and adjusted as required and the attitude control system will be tested. Solar arrays will be deployed and pointed sunward to verify the station's power generation systems. The station's communication system will be tested, ensuring connection to ground stations and orbital communication platforms alike. Once the life support systems have been tested and the internal environment has been verified as being ready for occupancy, the go ahead for crew launch will be given.

### *Crew Launch and Arrival*

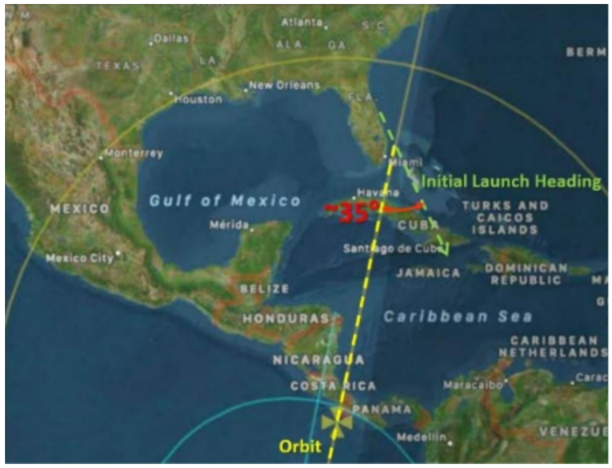


*Figure 3333. SpaceX's Fram2 launch trajectory to polar orbit from Kennedy Space Center (KSC) (David, 2025)*

The crew will begin final preparations for launch following verification that the station is ready for occupancy. The crew will be launched from one of the complexes on the Eastern Test Range on a similar trajectory to the one used by Fram2, shown above in Figure 1. Following orbital insertion, the crew will spend time cruising and performing orbital adjustments before rendezvous with Boyd Station. Once the crewed spacecraft enters proximity and station keeping of the space station, it will perform a docking procedure using visual aids mounted on the exterior and docking radar data. The spacecraft will approach slowly and once established in the right attitude, dock with the station. Once the cabin environment has equalized with the habitat environment on the station, the crew will open hatch between the two and ingress the station.

Following ingress, the crew will begin performing a checkout of the interior of the station. This will include properly installing and turning on the Environmental Control and Life Support System (ECLSS), C3, or science systems not prepared during orbital checkout or ground closeout. The first few hours on board will also include verifying the station is habitable and loading equipment, supplies and personal items from the spacecraft into the hab. The final step of ingress will be shutting down the visiting spacecraft's systems and integrating them with the station: accommodation for power, thermal, data, and ECLSS.

## On-Orbit Operations

### *Crew Habitation*

The crew will stay on board the station for a total of seven days. In that time they will be performing science experiments, working on station upkeep, and preparing for an EVA that is scheduled to occur between the fourth and sixth day of the mission. The crew will be able to sleep in their private crew quarters, enjoy meals in the shared galley space, and observe Earth through windows during personal time. The greater details of regular habitation will be covered in the Operations Concept.

### *On-Orbit Research*

Boyd Station's unique orbit provides a wide range of scientific goals due to the polar orbital environment. A SSO gives the station access to high active portions of the ionosphere, magnetosphere, and radiation belt. This allows the station to be fitted with sensors to monitor these environments, even when the crew is not on board. Passing through the radiation belts allows the crew and biological samples brought with them to be exposed to limited dosages of radiation. This can simulate hazardous regions of space like the proper Van Allen belts, and can serve as an analog for dosages on long term missions without the risk of actual long term exposure. Additionally, if hazardous levels of radiation are detected, the crew can safely depart the station before they receive too high a dosage.

Boyd Station is fitted with accommodations for four active Middeck Locker Equivalents, allowing crewed missions to have high configurability for experiments. The station's environment also allows for contact with many places over the Earth's surface. Crews can test remote sensing and communications hardware over a number of surface environments, and configure them in orbit to get the best results. These results can be compared from day to day as the SSO orbit allows for viewing of sites at the same solar time every day. There are opportunities for polar and boreal Earth observation that cannot be achieved by the ISS.

### *EVA Operations*

Between the fourth and sixth days of the mission on board Boyd Station, two of the three crew members will perform an EVA outside of the station. The spacewalk will occur near the end of the stay to ensure the crew have as much time as possible to acclimatize to the pressure and atmosphere needed in the spacesuits. While their life support will be provided for by the station through an umbilical, the suits needed for the spacewalk will be provided by the visiting crew. To reduce launch mass of the station, it will not be equipped with an airlock and the third crewmember will wear an EVA suit inside the habitat. A large portion of time spent on the day of the EVA will be preparing the habitat for atmosphere evacuation, and getting it set back up following depressurization. The station will include a set of tether points and hand rails to ensure the team can safely navigate the exterior during the spacewalk.

### *Crew Egress and Departure*

Following the EVA, the primary focus of the mission will be on closing out the station prior to crew departure. This will include gathering waste products and properly disposing of them, either by jettisoning solid waste through the trash airlock or performing a waste water/waste gas dump overboard. Science experiments will be closed out and shut down, and any materials or data that need to be returned with the crew will be stored on the visiting spacecraft. Similarly, personal effects and other supplies needing to be returned will be loaded before the station is placed in a dormant state. The visiting crewed spacecraft will then disconnect all of the umbilical connections with the station, close the hatch and undock. After a slow departure from the vicinity of Boyd Station, the crewed spacecraft will be able to perform a deorbit burn and return to Earth.

## Uninhabited and Autonomous Operations

As the station is only planned for ten week-long stays during a five-year period, there will be large periods of time when the station will need to fly autonomously. During this time it will be monitored by a ground crew to maintain optimal attitude for station keeping and solar collection, and to ensure command and control systems remain operational. With the station uninhabited, many major power-intensive functions will be left unpowered, providing the station a chance to orbit without perfect solar incidence to allow for a reduced orbital decay rate. Orbital altitude maintenance will be a major source of required propellant mass, so decay reduction during uninhabited flight will be critical to station mass reduction. Autonomous monitoring equipment supporting secondary science missions will be able to download data to the ground.

## Follow On Crewed Habitation

Follow on crews will continue to arrive and depart on a similar trajectory as the original crewed mission. The station is designed to meet its own power, thermal, and propulsion needs for the entire five-year mission, all missions after the first will need to resupply Boyd station prior to crew arrival. During the first few hours on board the station, it will be the crew's job to restock the atmosphere gas storage, water tanks, food stores and other consumables from their spacecraft. Crews that use the station later in its life can expect to spend more time on station upkeep during their weeklong stay. As hardware degrades over the course of the five-year mission, more time will be dedicated to ensuring critical systems like ECLSS remain operational. This might require crews to perform preventive or corrective maintenance during their stay.

## Station Decommission and Demise

At the conclusion of the station's five-year lifetime, it will be made ready for decommissioning by the final visiting crew. This will include removing any hardware or experiments that need to have materials processed on the ground, securing waste and excess consumables in the best location for burn up, and securing any pressurized liquids or gasses while the station is in an autonomous mode. When the crew departs, the station will be in an autonomous operation mode and prepared for deorbit.

Using the remaining onboard propellant, the station will lower its orbit until it is low enough for a stable demise (around ~250 km). Once the station is in that lower orbit, it will perform one final orbit lowering burn to aim the descent at the southern Pacific Ocean (periapsis lowered to 90 km). The station will also place itself in a high drag attitude configuration, facing the long axis of the station along with the solar panels into the airstream to ensure that as much of the station will burn up during the descent as possible. This configuration ensures as little debris will make it to the surface as possible, and that any that survives reentry will harmlessly fall into the ocean.

## Concept of Operations (ConOps)

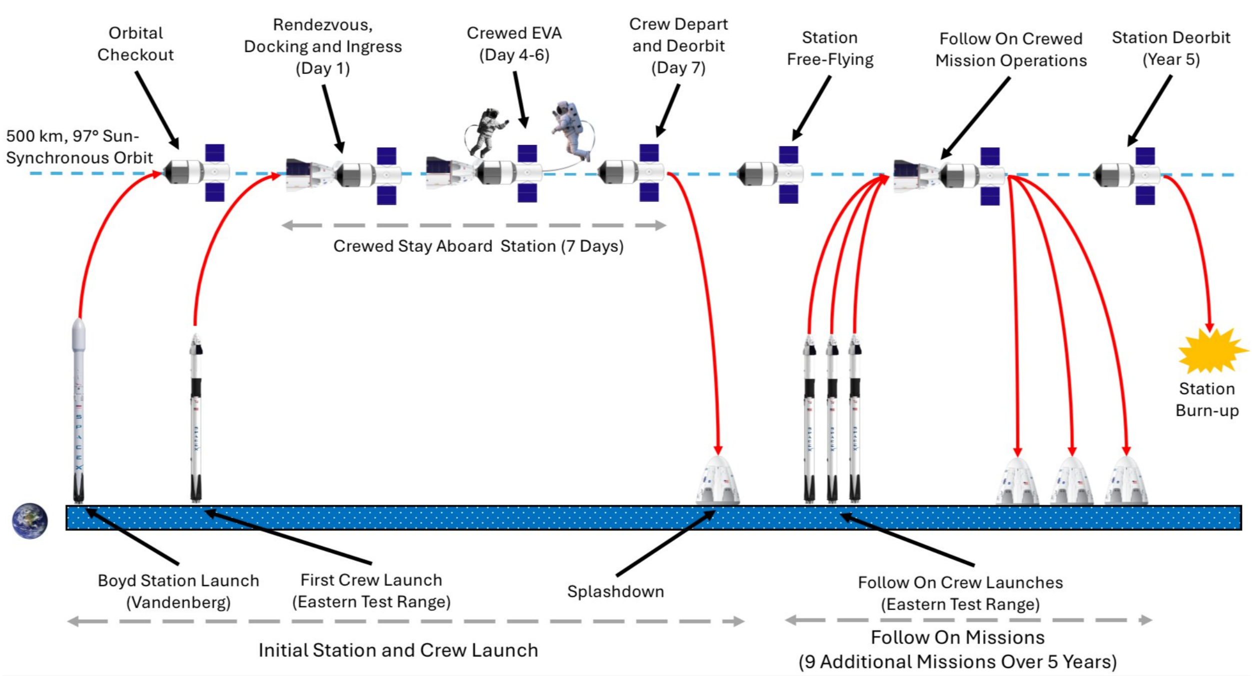


*Figure 4444. Boyd Station ConOps*

## Functional Decomposition (FD)

The functional decomposition process – which can be observed in the FD excel file submitted separately – allowed the team to break down high-level functional objectives into low level requirements that could be met with various technologies.

The high-level functional objectives were originally derived from the ground rules and mission goals. Three humans in the equation yielded the addition of the most critical requirements – keeping the crew alive. In fact, a significant portion of the requirements derived from the FD are dedicated to the ECLSS and CA subsystems, which revolve around the crew's health and well-being.

Figure 5 represents a sample FD breakdown for the high-level FO stating that "the station shall maintain nominal operation". For simplicity, the figure only shows the full decomposition process for "power distribution" which is a level 2 function that was identified. The resulting technologies have more in-depth features (M,P,V properties) called out in FD as well and are not shown in Figure 5.

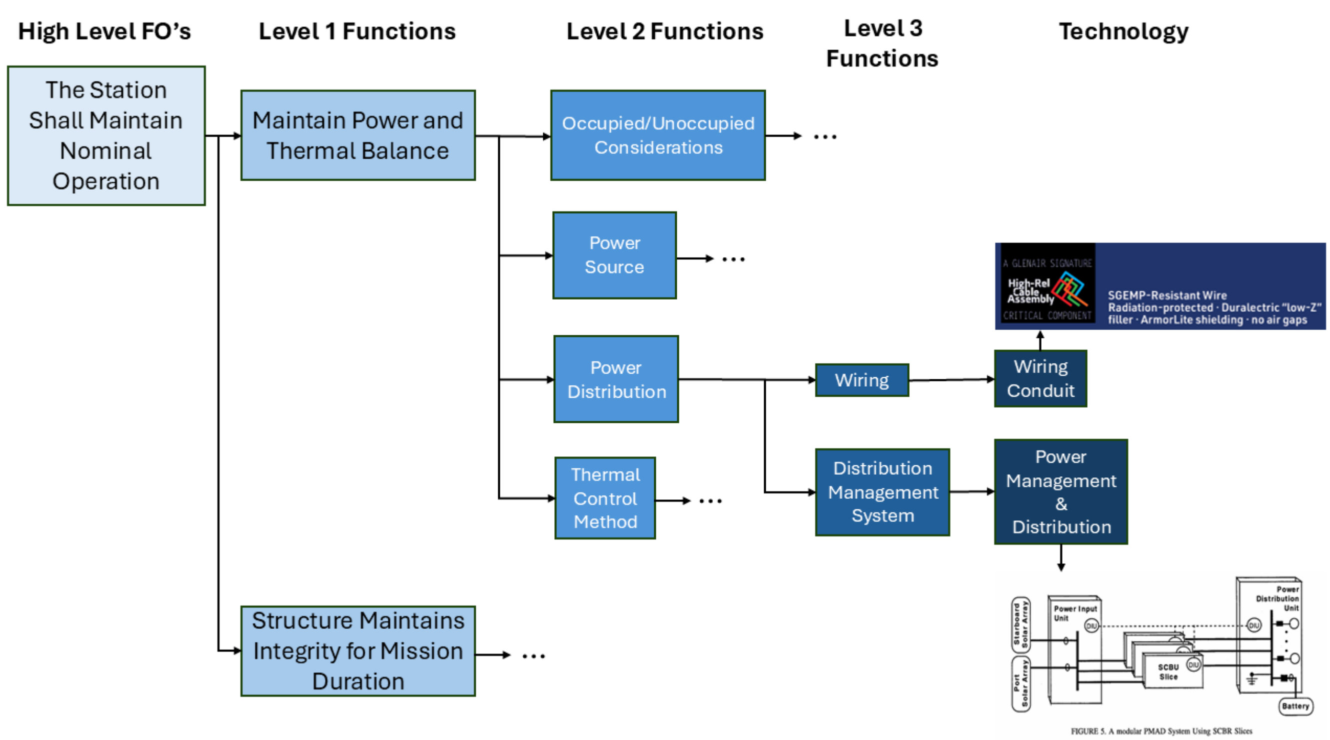


*Figure 5555. FD Breakdown Sample.*

Utilizing the FD spreadsheet allowed the team to identify the technology and equipment needed to meet the needs of a mission and provided an organized way to track TRL, mass, power, and volume specifications.

## Operational Concept (OpsCon)

The day-to-day operations of the mission are summarized in the tables below. Some activities only occur on days where no EVA is performed, and vice versa. Leisure time can also be reduced if other activities exceed their time. The timing of many of the nominal activities are based on an example nominal day's OpsCon aboard the ISS (ESA, 2014). The required EVA pre-breathe time is based on the habitat's and spacesuit's atmospheres (McBarron, 1993). The preparation and conclusion activities were timed based on a presentation by the United Space Alliance and based on NASA procedures (Moore & Marmolejo, 2013).

Daily Activities account for 16.5 hours of the day while EVA Activities and Nominal Day Activities each count for 7.5 hours, adding up to a total of 24 hours. If the crew goes to bed at 9:30 pm and wakes at 6:00 am as they do on the ISS (ESA, 2014), it means the crew should arrive to the habitat by 12:30 pm on the arrival day and leave at about 4:00 pm on the final day. Some activities will not occur on the arrival and departure days due to the timing (for example, the evening meal will not occur on the habitat on the final day). Arrival procedures include transferring consumables into the habitat and ensuring the habitat is in working order. Departure procedures include housekeeping duties, final collection of trash, collection of mission-specific items that are returning to Earth, and any other shutdown procedures that should be done before leaving the habitat.

### Daily Activities (All Crew)

*Table 2222. Daily Activities OpsCon.*

| Crew Activity | Hours Needed | Equipment |
|---|---|---|
| Sleep | 8.5 | Crew Quarters, Personal Preference Kit (PPK) (ESA Blogteam, 2014) |
| Pre-sleep | 1.5 | PPK (ESA Blogteam, 2014) |
| Post-sleep | 1.5 | PPK (ESA Blogteam, 2014) |

| | | |
|---|---|---|
| Personal Hygiene (PH) (morning) | 0.5 | Personal Hygiene Kit (PHK), PH Consumables |
| PH (evening) | 0.5 | PHK, PH Consumables |
| Morning meal | 0.5 | Galley, Food, Utensils (ESA Blogteam, 2014) |
| Midday meal | 1 | Galley, Food, Utensils (ESA Blogteam, 2014) |
| Evening meal | 0.5 | Galley, Food, Utensils (ESA Blogteam, 2014) |
| Communications | 2 | Habitat Computer |

## EVA Activities (Two Crew Members)

*Table 3333. EVA Activities OpsCon*

| Crew Activity | Hours Needed | Crew Involved | Equipment |
|---|---|---|---|
| Preparation | 1.5 | 3 | EVA Suit (Moore, 2013) |
| Pre-breathe | 0.6667 | 3 | EVA Suit, Umbilical (McBarron, 1993) |
| EVA | 3 | 2 | EVA Suit, Umbilical |
| In-capsule | 3 | 1 | EVA Suit, Umbilical |
| Conclusion | 1.5 | 3 | EVA Suit (Moore, 2013) |
| Leisure | 0.8333 | 3 | PPK |

### Nominal Day Activities (All Crew)

*Table 4444. Nominal Day Activities OpsCon.*

| Crew Activity | Hours Needed | Equipment |
|---|---|---|
| ECLSS Maintenance | 0.5 | Replacement Parts, Tools |
| Habitat Maintenance | 1 | Replacement Parts, Tools |
| Experiments | 3 | Middeck Locker Equivalent |
| Leisure | 2 | PPK |
| Photography | 1 | Cameras |

### Once per Mission Activities (All Crew)

*Table 5555. Single Occurrence Activities OpsCon.*

| Crew Activity | Hours Needed | Equipment |
|---|---|---|
| Arrival Procedures | 3 | PPK, Clothes Bag, Replacement Parts, EVA Suit |
| Departure Procedures | 3 | PPK, Clothes Bag, EVA Suit |
| Waste Jettison | 1 | Waste System |

# Conceptual Design Verification

## Subsystem Definitions, Requirements and Technologies

The Functional Decomposition is a breakdown of the high-level Functional Objectives to first determine the main functions of the subsystems. Then the Functional Decomposition breaks these high-level subsystem requirements to the smallest level where there is an associated technology that can fulfill each requirement. The Operational Concept is a list of the activities that the crew performs during the 7-day mission. Each activity is related to at least one subsystem and piece of equipment.

## ECLSS

The high-level function of ECLSS is to keep the crew alive throughout the mission. To achieve this, ECLSS must regulate the atmosphere, supply food, manage waste, and manage water. These four primary functions are represented by the blue corner blocks in the ECLSS Functional Block Diagram (Figure 6). The corresponding lowest-level requirements and the selected technologies for each requirement are detailed in Table 6 and Table 7, respectively. Importantly, the short mission duration (7 days) has been a key driver in the selection and simplification of the required ECLSS technologies.

A total cabin pressure of 10.2 psia was selected for Boyd Station to simplify prebreathe protocols, mirroring the reduced-pressure atmosphere used by the Space Shuttle during EVA preparation (NASA Exploration Atmospheres Working Group, 2010).

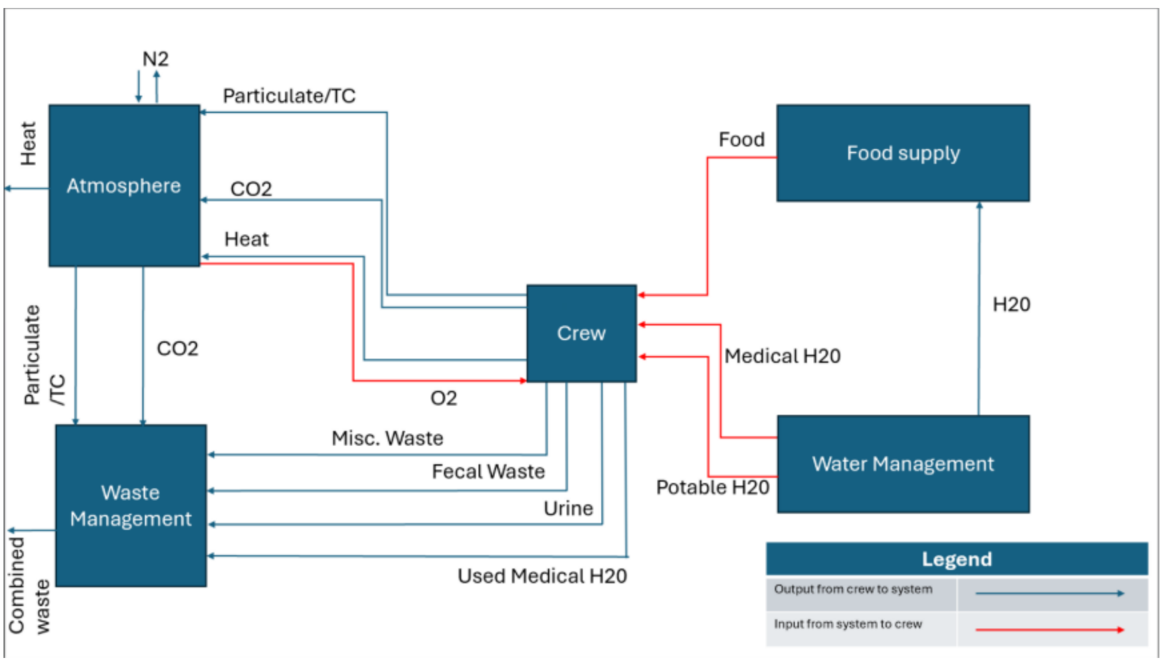


*Figure 6666. ECLSS system interface Functional Block Diagram (FBD).*

*Table 6666. ECLSS Functional Objectives (FOs).*

| **ID** | **Requirement Statement** | **Rationale** | **Verification Method** |
|---|---|---|---|
| ECLSS-01 | The ECLSS system shall provide and continuously control the cabin $O_2$ partial pressure at 2.46 psia (24.11%) | Crew Safety (NASA Exploration Atmospheres Working Group, 2010) | Test |

| | | | |
|---|---|---|---|
| ECLSS-02 | The ECLSS system shall provide and continuously control the cabin $N_2$ partial pressure at 7.74 psia (75.89%) | Crew Safety (NASA Exploration Atmospheres Working Group, 2010) | Test |
| ECLSS-03 | The ECLSS system shall provide circulation/ventilation through mono-directional flow | Crew Safety | Test |
| ECLSS-04 | The ECLSS system shall keep RH between 30% and 70% | ISS Nominal Values (NASA) | Test |
| ECLSS-05 | The ECLSS system shall keep $ppCO_2$ below 3 mmHg | National Institute for Occupational Safety and Health (NIOSH) and National Aeronautics and Space Administration (NASA) Nominal Values (NASA, 2025) | Test |
| ECLSS-06 | The ECLSS system shall keep particulates below 3 mg/m3 | NASA Nominal Value (NASA, 2025) | Test |
| ECLSS-07 | The ECLSS system shall keep VOCs and Trace Contaminates (TC) below SMAC defined levels | NASA Nominal Values (NASA, 2024) | Test |
| ECLSS-08 | The ECLSS system shall keep bacteria in air below 1000 CFU/m3 | NASA Nominal Values (NASA, 2025) | Test |
| ECLSS-09 | The ECLSS system shall keep fungi in air below 100 CFU/m3 | NASA Nominal Values (NASA, 2025) | Test |
| ECLSS-10 | The ECLSS system shall monitor temperature | Crew Safety | Analysis |
| ECLSS-11 | The ECLSS system shall monitor humidity | Crew Safety | Analysis |

| | | | |
|---|---|---|---|
| ECLSS-12 | The ECLSS system shall monitor $CO_2$ | Crew Safety | Analysis |
| ECLSS-13 | The ECLSS system shall monitor total pressure | Crew Safety | Analysis |
| ECLSS-14 | The ECLSS system shall monitor partial pressures | Crew Safety | Analysis |
| ECLSS-15 | The ECLSS system shall monitor hazardous contamina nts | Crew Safety | Analysis |
| ECLSS-16 | The ECLSS system shall monitor fire byproducts (CO, $CO_2$, Energy) | Crew Safety | Analysis |
| ECLSS-17 | The ECLSS system shall monitor biological contaminan ts | Crew Safety | Analysis |
| ECLSS-18 | The ECLSS system shall send information to control subsystems and crew alert system | Crew Safety | Test |
| ECLSS-19 | The ECLSS system shall provide fire suppression equipment | Crew Safety | Inspection |
| ECLSS-20 | The ECLSS system shall provide and store sterile 2.5 kg/CM-d (kg/crew member per day) Potable water | Life Support Baseline Values and Assumptions Document (BVAD) Values | Inspection |
| ECLSS-21 | The ECLSS system shall store sterile and provide 5 + 0.5 kg/CM-d Medical water(15.5 kg total) | BVAD Values | Inspection |
| ECLSS-22 | The ECLSS system shall collect and Store Human Derived Waste | Crew Safety | Inspection |
| ECLSS-23 | The ECLSS system shall jettison Waste | Ground Rule | Test |

| | | | |
|---|---|---|---|
| ECLSS-24 | The ECLSS system shall provide 2.39 kg/CM-d of dry food (weight includes packaging) | BVAD Values | Inspection |

*Table 7777. ECLSS Master Equipment List (MEL).*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume ($m^3$) |
|---|---|---|---|---|---|
| ECLSS-01.1 | Pressurized $O_2$ Tank | 1 | 42 | 0 | 0.034 |
| ECLSS-01.2 | $O_2$ | 1 | 29 | 0 | 1.700 |
| ECLSS-01.3/02.3 | Vent/Relief Valve | 2 | 10.8 | 0 | 0.028 |
| ECLSS-01.4/02.4 | Positive Pressure Relief Valve | 2 | 1.8 | 0 | 0.003 |
| ECLSS-01.5/02.5 | Negative Pressure Relief Valve | 2 | 2 | 0 | 0.006 |
| ECLSS-01.6/02.6 | General-Purpose Valves | 10 | 11 | 0 | 0.200 |
| ECLSS-01.7/02.7 | Regulator | 2 | 3 | 0 | 0.004 |
| ECLSS-02.1 | Pressurized $N_2$ Tank | 1 | 19 | 0 | 0.273 |
| ECLSS-02.2 | $N_2$ | 1 | 10 | 0 | 0.650 |
| ECLSS-03 | IMV Fan | 2 | 9 | 110 | 0.019 |
| ECLSS-04 | Condensing Heat Exchanger System | 1 | 22 | 92 | 0.078 |
| ECLSS-05 | LiOH Canister | 21 | 76 | 0 | 0.026 |

| | | | | | |
|---|---|---|---|---|---|
| ECLSS-06/08/09 | HEPA Filter | 3 | 6 | 0 | 0.025 |
| ECLSS-07 | Catalytic Filter | 2 | 11 | 0 | 0.016 |
| ECLSS-07 | Activated Charcoal | 2 | 74 | 0 | 0.153 |
| ECLSS-10 | Temperature sensor | 4 | 1 | < 1 | 0.002 |
| ECLSS-11 | Humidity/liquid sensor | 2 | 1 | < 1 | 0.001 |
| ECLSS-12 | $CO_2$ Sensor | 2 | < 1 | 1 | < 0.001 |
| ECLSS-13 | Total Cabin Pressure Sensor | 2 | < 1 | 1 | 0.006 |
| ECLSS-14/15 | Mass spectrometer | 1 | 14 | 34 | 0.024 |
| ECLSS-16 | Fire Detection Assembly | 1 | 2 | 1 | 0.003 |
| ECLSS-17 | Bacteria Filter Assembly | 1 | 5 | 0 | 0.122 |
| ECLSS-19.1 | $CO_2$ Fire Extinguisher | 1 | 5 | 0 | 0.041 |
| ECLSS-19.2 | Emergency Respirator | 3 | 20 | 0 | 0.030 |
| ECLSS-20.1 | Potable Water | 1 | 80 | 0 | 1.632 |
| ECLSS-20.2 | Potable Water Tank | 1 | 16 | 0 | 0.002 |
| ECLSS-20.3/21.3 | Nitrogen Pump for Water Tanks | 2 | 5 | 0 | < 0.001 |
| ECLSS-21.1 | Medical Water Tank | 1 | 5 | 0 | 0.001 |

| | | | | | |
|---|---|---|---|---|---|
| ECLSS-21.2 | Medical Water | 1 | 26 | 0 | 0.026 |
| ECLSS-23.1 | Trash Airlock | 1 | 150 | 0 | 0.300 |
| ECLSS-23.2 | Fecal Canisters | 3 | 7 | 0 | 0.041 |
| ECLSS-24.1, CA-09.1 | Pre-Packaged Food (served ISS pantry style) | 21 | 41 | 0 | 0.132 |
| ECLSS-24.2, CA-09.2 | Food Packaging | 21 | 9 | 0 | 0 |
| ECLSS-24.3, CA-09.3 | Margin for Food | 6 | 12 | 0 | 0.038 |
| ECLSS-24.4, CA-09.4 | Packaging for Margin | 6 | 3 | 0 | 0 |

## CA/PA/EVA

The EVA/CA/PA subsystem is responsible for determining the overall accommodation for the crew, payload, and procedure for conducting the EVA. This encapsulates keeping the crew happy and productive during their time aboard the habitat and maintaining the payloads at a functional working condition. The block diagram in Figure 7 separates the inputs and outputs by CA, PA, and EVA. The requirements of each subsystem are in Table 8 and the associated equipment is in Table 9. Some requirements and equipment overlap with ECLSS and are in Tables 6 and 7.

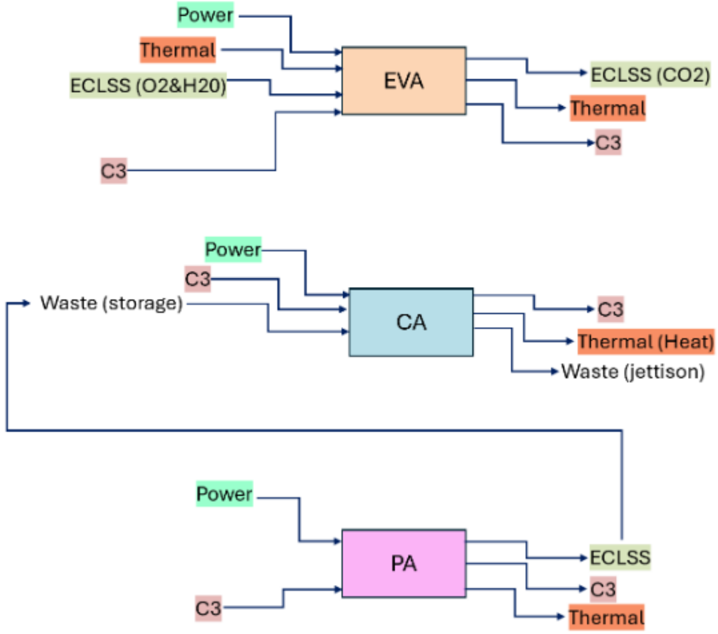


*Figure 7777. CA/PA/EVA system interface FBD.*

*Table 8888. Table 10. EVA/CA/PA FOs.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|
| EVA-01 | The EVA suit shall operate at 4.3 psia | Suit spec | Test |
| EVA-02 | The EVA suit shall provide 100% $O_2$ | Suit spec | Test |
| EVA-03 | An umbilical system shall be provided for EVA | Ground rule | Inspection |
| EVA-04 | The EVA shall last for 3 hours | Ground rule | Demonstration |
| EVA-05 | The crew shall perform a 40-minute pre-breathe procedure prior to EVA | Crew safety | Demonstration |
| CA-01 | The crew shall have hygiene equipment onboard the habitat | Crew comfort | Inspection |

| | | | |
|---|---|---|---|
| CA-02 | The crew quarters shall incorporate ventilation and acoustic control | Crew comfort | Test |
| CA-03 | The crew quarters shall include wall attachment hooks and sleeping bags | Crew comfort | Inspection |
| CA-04 | The crew shall have cleaning equipment onboard the habitat | Crew safety | Inspection |
| CA-05 | The crew shall have access to health care onboard the habitat | Crew safety | Inspection |
| CA-06 | The crew shall have tools to perform onboard maintenance on the habitat | Crew safety | Inspection |
| CA-07 | The crew shall have restraints and mobility aids for maneuvering around the habitat | Crew safety | Inspection |
| CA-08 | The crew shall have a location for disposing of non-human waste | Crew safety | Inspection |
| CA-09 | The crew shall have adequate storage for food. | Crew comfort | Inspection |
| CA-10 | The crew shall have a rehydration unit for meals | Crew safety | Test |
| CA-11 | The crew shall be allowed 1 PPK (Personal Preference Kit) for approved personal items of choice | Crew comfort | Inspection |
| CA-12 | The crew shall have a functioning toilet with contingency human waste bags | Crew safety | Test |
| CA-13 | The crew shall have clothing onboard with adequate storage | Crew comfort | Inspection |
| CA-14 | The habitat shall have sufficient lighting in convenient locations | Crew Safety | Test |
| CA-15 | The habitat shall provide a private sleeping space for each crew member | Crew Comfort | Inspection |

| | | | |
|---|---|---|---|
| PA-01 | The habitat will have cameras in the common area for crew observation | Mission Goal | Test |
| PA-02 | The habitat shall have 4 MLE for experiments onboard | Mission Goal | Test |
| PA-03 | The habitat shall have functional remote sensing instrumentation for radiation and climatological research | Mission Goal | Test |
| PA-04 | Appropriate R&MA shall be provided to allow crew to access remote sensing equipment outside of the spacecraft | Crew safety | Inspection |
| PA-05 | The habitat will have equipment for polar-orbit-focused human and biological experiments | Mission Goal | Test |

*Table 9999. EVA/CA/PA MEL.*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|---|---|
| EVA-03 | Umbilical | 2 | 7 | 308 | 0.005 |
| EVA-04 | Mobility Aids | 1 | 40 | 0 | 0.260 |
| EVA-02 | LiOH Canister (EVA Specific) | 3 | 41 | 0 | 0.017 |
| CA-01.1 | Personal Hygiene Kit | 21 | 2 | 0 | 0.015 |
| CA-01.2 | Hygiene Supplies (Consumables) | 3 | 1 | 0 | 0.032 |
| CA-02 | Ventilation and Acoustic Design for CQ | 3 | 23 | 33 | 0.825 |

| | | | | | |
|---|---|---|---|---|---|
| CA-03.1/CA-08.1 | Magnetic Carabiner | 25 | <1 | 0 | <0.001 |
| CA-03.2 | ISS Sleeping Bag | 3 | 3 | 0 | 0.110 |
| CA-04.1 | Kitchen Cleaning Supplies/Trash | 7 | <1 | 0 | 0.013 |
| CA-04.2 | Disposable Wipes | 21 | <1 | 0 | 0.882 |
| CA-05 | Medical Aid Kit | 1 | 7 | 0 | 0.150 |
| CA-06 | In-Flight Tool-Kit | 1 | 38 | 0 | 0.200 |
| CA-07 | Restraints and Mobility Aids (R&MA) | 1 | 83 | 0 | 0.540 |
| CA-08.2 | Trash Bags | 42 | <1 | 0 | 0.042 |
| CA-10.1 | Food Rehydration Unit | 1 | 15 | 200 | 0.014 |
| CA-10.2 | Hot Pads | 3 | <1 | 0 | 0.003 |
| CA-11 | Personal Preference Kit | 3 | 1 | 0 | 0.004 |
| CA-12.1, ECLSS-22 | Space Toilet | 1 | 45 | 45 | 2.180 |
| CA-12.2 | Contingency Human Waste Bags | 21 | <1 | 0 | 0.006 |
| CA-13.1 | Clothes | 21 | 2 | 0 | 0.168 |
| CA-13.2 | Clothes Storage | 3 | <1 | 0 | 0.183 |
| CA-14 | Internal General Lighting Assembly | 1 | 4 | 30 | 0.010 |
| CA-15 | ISS Crew Quarters | 3 | 360 | 0 | 6.300 |

| | | | | | |
|---|---|---|---|---|---|
| PA-01 | Cameras for Observation | 3 | 1 | 0 | 0.004 |
| PA-02 | Middeck Locker Payloads | 4 | 100 | 1000 | 0.272 |
| PA-03 | Remote Sensing Instrumentation | 5 | 1 | 2 | 0.001 |

## Avionics/C3

The avionics subsystem provides navigation controls for the habitat while it is in orbit, and reliable communication to the ground throughout the duration of the habitat's lifetime.

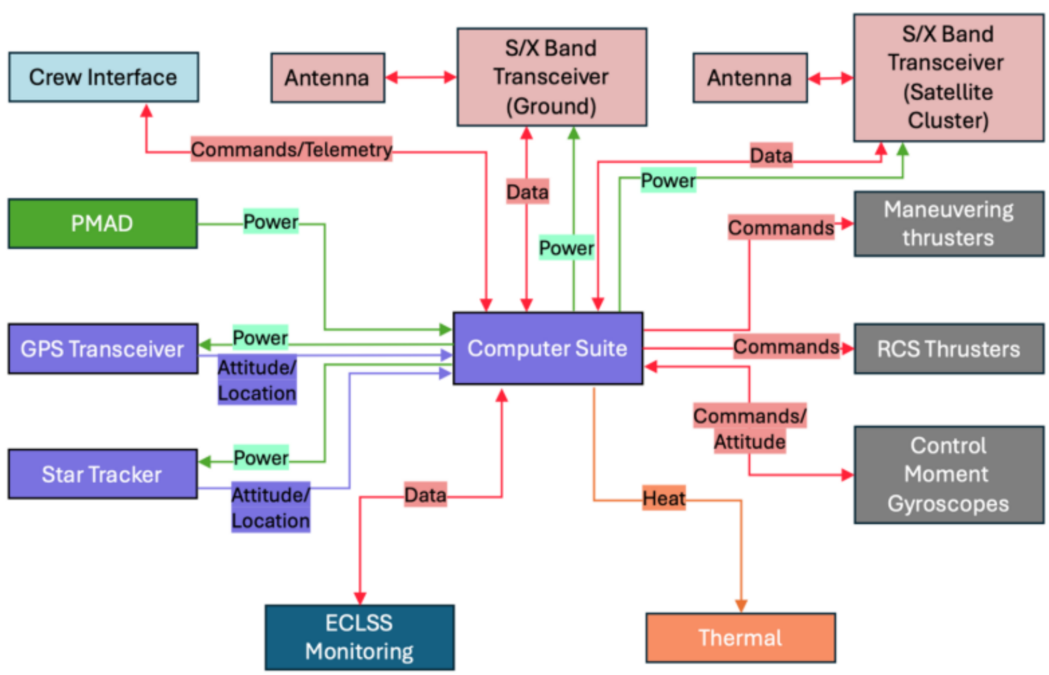


*Figure 8888. Avionics/C3 system interface FBD.*

*Table 10101010. Avionics/C3 FOs.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|

| | | | |
|---|---|---|---|
| AVN-01 | The station shall be able to command attitude inputs to GNC subsystem | Avionics Requirement | Test |
| AVN-02 | The station will calculate orbital adjustments as necessary | Avionics Requirement | Test |
| AVN-03 | The station shall have the ability to calculate where it is in space and orbit | Avionics Requirement | Test |
| C3-01 | The station shall facilitate constant two-way communication and data transfer with ground control | Communications Requirement | Test |
| C3-02 | The station shall allow command inputs from astronauts on board | Crew Safety | Test |

*Table 11111111. Avionics/C3 MEL.*

| **Equipment ID** | **Equipment Name** | **Quantity** | **Total Mass (kg)** | **Total Power (W)** | **Total Volume ($m^3$)** |
|---|---|---|---|---|---|
| AV-01, ECLSS-18 | Onboard Computer | 2 | 19 | 90 | 0.022 |
| AV-03.1 | GNSS Receiver | 2 | 2 | 0 | 0.005 |
| AV-03.2 | Star Tracker | 3 | 1 | 6 | 0.001 |
| C3-01.1 | Ku-Band Antenna | 2 | < 1 | 0 | 0.001 |
| C3-01.2 | S-Band Antenna | 2 | 1 | 0 | 0.002 |
| C3-01.3 | X-Band Antenna | 2 | 1 | 0 | 0.003 |

| | | | | | |
|---|---|---|---|---|---|
| C3-01.4 | Ku-Band Transceiver | 1 | 1 | 6 | < 0.001 |
| C3-01.5 | S/X Band Transceiver | 1 | 4 | 8 | 0.004 |

## Power

The power subsystem is vital for the nominal operation of the station, as all subsystems outside of structures require a set amount of power to function. To achieve this, the power subsystem must generate power to meet subsystem demands, provide adequate storage for when power is not actively being generated, and condition and distribute stable power throughout the station. These functions are represented by the functional block diagram in Figure 9, with power generated by solar arrays being conditioned to be sent to the relevant subsystems and battery storage. Tables 12 and 13 describe the lowest level requirements of the power system and the associated equipment to meet these requirements.

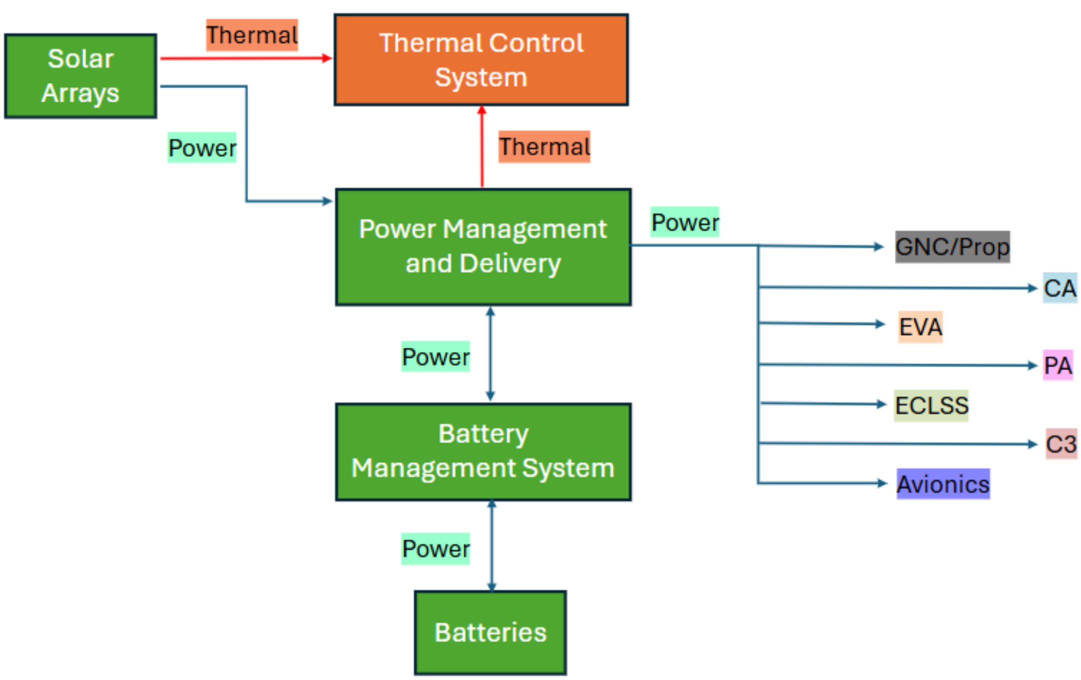


*Figure 9999. Power system interface FBD.*

*Table 12121212. Power FOs.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|
| PWR-01 | The Power system shall generate solar power to meet end-of-life requirements | Nominal operation | Analysis |
| PWR-02 | The Power system shall convert and condition generated power (DC/DC, AC inversion) | Nominal operation | Test |
| PWR-03 | The Power system shall provide regulated bus voltages (primary and secondary) | Nominal operation | Test |
| PWR-04 | The Power system shall distribute power to meet energy demands for occupied/dormant conditions | Nominal operation | Test |
| PWR-05 | The Power system shall provide adequate energy storage for operation in loss of sunlight | Nominal operation | Test |
| PWR-06 | The Power system shall provide fault isolation | Crew safety | Test |
| PWR-07 | The Power system shall maintain battery health during charge/discharge cycles (over current, temperature limits) | Nominal operation | Test |

| | | | |
|---|---|---|---|
| PWR-08 | The Power system shall provide redundancy and safe-mode power configurations for critical life support/emergency systems | Crew safety | Test |

*Table 13131313. Power MEL.*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|---|---|
| PWR-01 | Solar Arrays | 2 | 49 | 0 | 0.628 |
| PWR-02 | Power Management & Distribution | 2 | 12 | 0 | 0.026 |
| PWR-03 | Li-Ion Batteries | 3 | 6 | 0 | 0.12 |
| PWR-04 | Wiring | 200 | 24 | 0 | 2.76 |

## Thermal

The thermal subsystem is responsible for maintaining operable temperatures for both the crew and equipment. This allows the crew to remain healthy and productive throughout the duration of occupancy and to ensure nominal operation of the station. The function block diagram in Figure 10 shows the thermal rejection loop of the station, with waste heat from the crew, subsystems, and electronics being expelled to the external space environment. The Functional Objectives for the thermal subsystem are given in Table 10. The technologies required to meet the objectives are listed in Table 11.

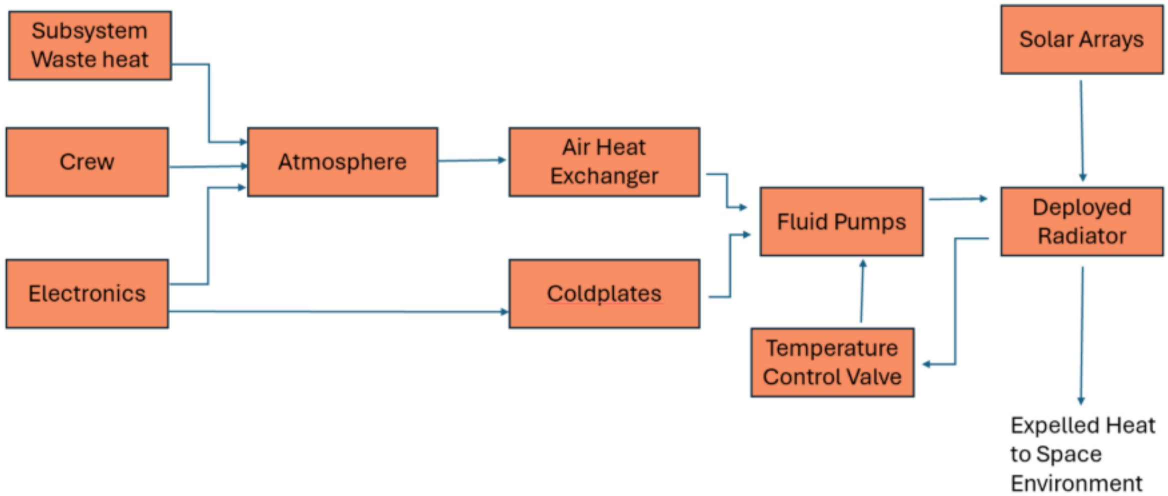


*Figure 10101010. Thermal system interface FBD.*

*Table 14141414. Thermal FOs.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|
| THM-01 | The Thermal system shall maintain cabin temperature range of 18°C-25.7°C while crew is onboard | Crew safety | Test |
| THM-02 | The Thermal system shall maintain operable temperature for equipment during launch prep and ascent | Nominal operation | Test |
| THM-03 | The Thermal system shall maintain operable temperature for equipment during nominal operation | Crew safety | Test |
| THM-04 | The Thermal system shall maintain crew accessible surface temperatures below 40°C | Crew safety | Test |
| THM-05 | The Thermal system shall provide adequate thermal storage to buffer temperature swings during orbit | Crew safety/nominal operation | Analysis |

| | | | |
|---|---|---|---|
| THM -06 | The Thermal system shall prevent cold-soak conditions below 10°C in crew-occupied zones | Crew safety | Test |

*Table 15151515. Thermal MEL.*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|---|---|
| THM-01 | Air heat exchangers | 2 | 36 | 0 | 0.04 |
| THM-02 | Coldplates | 5 | 4 | 0 | 0.4 |
| THM-03 | Fluid pumps | 1 | 23 | 165 | 0.05 |
| THM-04 | Radiators (deployable) | 1 | 204 | 0 | 5.04 |
| THM-05 | Radiators (fixed) | 1 | 318 | 0 | 2.04 |
| THM-06 | TCS control instruments | 2 | 82 | 10 | 0 |
| THM-07 | Fluid plumbing | 2 | 246 | 0 | 0 |
| THM-08 | Fluid (Water) | 1 | 41 | 0 | 0 |
| THM-09 | Plumbing Insulation | 2 | 41 | 0 | 0 |

## Structures

The structures subsystem determines and maintains the structural integrity of Boyd Station over the entire lifetime of the mission. Structures ultimately provides the overall total pressurized volume and total habitable volume of Boyd Station.


*Figure 11111111. Structures interface FBD.*

*Table 16161616. Structures Fos.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|
| STR-01 | The structure shall shield the crew from radiation | Crew safety | Analysis |
| STR-02 | The structure shall be designed to withstand expected MMOD | Crew safety | Analysis |
| STR-03 | The pressure vessel shall withstand a pressure differential of 10.2 psia in accordance with ECLSS-01 and ECLSS-02 | Crew Safety | Analysis |
| STR-04 | The structure shall withstand pressure cycling in accordance with EVA procedures | Crew Safety | Analysis |
| STR-05 | The structure shall accommodate a bare minimum of 4 MLEs | Ground Rule | Analysis |
| STR-06 | The structure shall provide sufficient, semi-private sleeping quarters volume for 3 crew members | BVAD (NASA, 2022) | Analysis |
| STR-07 | The structure shall provide sufficient pressurized volume for stored commodities | Ground Rule | Analysis |
| STR-08 | The structure shall provide sufficient habitable volume for working lab area | Mission Objective | Analysis |
| STR-09 (THM-01) | The structure shall thermally insulate the pressurized volume to maintain homeostasis | Crew Safety | Analysis |

| | | | |
|---|---|---|---|
| STR-10 | The structure shall withstand the vibrational loads associated with launch and deployment | Crew Safety | Analysis |
| STR-11 | The structure shall provide a docking interface for a visiting crewed spacecraft | Mission Objective | Analysis |
| STR-12 | The structure shall provide a means of egress for EVA | Mission Objective | Analysis |

*Table 17171717. Structures MEL.*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|---|---|
| STR-01 | International Docking Adapter | 1 | 526 | 0 | 2.212 |
| STR-02 | Habitat Bare Structure | 1 | 1481 | 0 | 0.536 |
| STR-03 | Thermal Blanket | 1 | 126 | 0 | 1.066 |
| STR-04 | MMOD Shield | 1 | 412 | 0 | 1.066 |

The MMOD Shield volume is assumed to be the same as the thermal blanket volume since the surface area is assumed to be the same as well as the thickness. The docking adapter volume listed is the full external volume.

## GNC/Prop

The Guidance, Navigation and Control/Propulsion subsystem is responsible for keeping the spacecraft in a stable orbit and with a stable attitude. This includes non-reactive station keeping in the form of Control Moment Gyros (CMGs) and a Reaction Control System (RCS) for attitude adjustments, CMG desaturation and orbit reboosting over the station's lifetime. This system also includes the necessary requirements for launch and orbit demise. The functions of orbit and position determination have been included with the C3 subsystem, including star trackers and GNSS receivers. This data along with telemetry and commands are passed to the GNC subsystem.

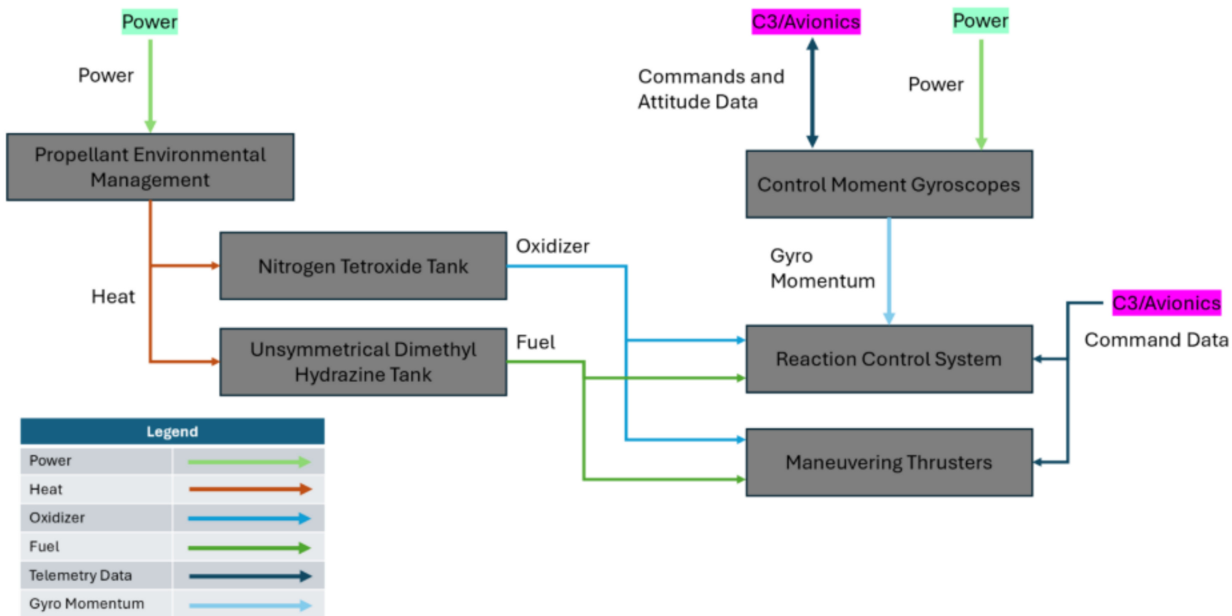


*Figure 12121212. GNC/Prop system interface FBD.*

*Table 18181818. GNC/Prop FOs.*

| ID | Requirement Statement | Rationale | Verification Method |
|---|---|---|---|
| GNCPROP-01 | The station shall be able to store propellant for the duration of the mission | Mission Goals | Test |
| GNCPROP-2 | The propulsion system shall be able to supply propellant to the population engines | Propulsion Requirements | Test |
| GNCPROP-03A | The station shall be able to perform attitude maintenance reactively | Ground Rules | Test |

| | | | |
|---|---|---|---|
| GNCPROP-03B | The station shall support rendezvous and docking procedures for a visiting crewed spacecraft | Mission Goals | Test |
| GNCPROP-03C | The station shall be able to perform contingency maneuvers to avoid MMOD hazards | Ground Rules | Test |
| GNCPROP-3D | The RCS system shall be able to maintain attitude and orbital altitude autonomously | Mission Goals | Test |
| GNCPROP-04A | The station shall perform a deorbit maneuver to ensure station demise at the end of the five-year mission | Mission Goals | Test |
| GNCPROP-04B | The station shall be able to correct its orbital parameters following launch | Mission Goals | Test |
| GNCPROP-05 | The station shall be able to maintain station keeping attitude non-reactively | Mission Goals | Test |
| GNCPROP-06 | The station shall launch into a Sun Synchronous Orbit | Ground Rules | Analysis |
| GNCPROP-07 (STR-10) | The station shall be able to withstand vibrational loads of launch | Structural Requirements | Test |
| GNCPROP-08 | The propulsion system shall be able to receive commands from the on-board C3 system and the ground | Ground Rules | Test |

*Table 19191919. GNC/Prop MEL.*

| Equipment ID | Equipment Name | Quantity | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| GNCPROP-01 | MMH Storage Tank | 2 | 19 | 12 | 0.46 |
| GNCPROP-02 | Prop Distribution | 1 | 22 | 0 | N/A |
| GNCPROP-03 | Attitude Control Thrusters | 1 | 177 | 345 | N/A |
| GNCPROP-04 | Reboost Thrusters | 1 | 191 | 112 | N/A |
| GNCPROP-05 | Propellant Mass | 1 | 2178 | 0 | N/A |
| GNCPROP-06 | Control Moment Gyros | 4 | 72 | 80 | 0.055 |

## Summary Subsystems MEL

The following MEL outlines the combined mass, power, and volume properties for each subsystem individually.

*Table 20202020. Total Subsystem MEL without margin.*

| Subsystem | Total Mass (kg) | Total Power (W) | Total Volume (m^3) |
|---|---|---|---|
| Avionics | 22 | 47 | 0.028 |
| C3 | 7 | 14 | 0.011 |
| CA | 1458 | 286 | 11.676 |
| ECLSS | 725 | 237 | 5.614 |
| EVA | 88 | 103 | 0.282 |
| GNC/Prop | 2336 | 549 | 1.14 |
| PA | 102 | 1050 | 0.277 |
| Power | 1352 | 0 | 3.534 |
| Structures | 2545 | 0 | 4.881 |
| Thermal | 994 | 165 | 7.57 |
| **Total:** | **9630** | **2451** | **35.013** |

# Subsystem Interfaces

## ECLSS

*Table 21212121. ECLSS quantitative Inputs and Outputs (I/Os).*

| Origin Subsystem | Input | Amount | Unit | Recipient Subsystem |
|---|---|---|---|---|
| ECLSS (Atmo.) | Heat | 237 | W | Thermal |

| | | | | |
|---|---|---|---|---|
| ECLSS (Atmo.) | $O_2$ | 1 | kg/CM-d | Crew |
| ECLSS (Atmo.) | $N_2$ | <1 | kg/CM-d | ECLSS (Atmo.) |
| ECLSS (Atmo.) | $CO_2$ | 1 | kg/CM-d | ECLSS (Waste) |
| ECLSS (Atmo.) | $O_2$ | <1 | kg/CM-d | EVA |
| ECLSS (Food) | Food | 41 | kg/C-m | Crew |
| ECLSS (Waste) | Combined human derived waste | 23 | kg/CM-d | Jettison |
| ECLSS (Water) | Potable $H_2O$ | 3 | kg/CM-d | Crew |
| ECLSS (Water) | Potable $H_2O$ | 1 | kg/CM-d | ECLSS (Food) |
| ECLSS (Water) | Medical $H_2O$ | 1 | kg/CM-d | Crew |

## CA/PA/EVA

*Table 22222222. CA/PA/EVA quantitative I/Os*

| Origin Subsystem | Input | Amount | Unit | Recipient Subsystem |
|---|---|---|---|---|
| CA | Heat | 108 | W | Thermal |
| CA | Non-human derived waste | 49 | kg/CM-d | Jettison |
| EVA | CO2 | 1 | kg/CM-d | ECLSS (Waste) |
| EVA | Heat | 3 | W | Thermal |
| PA | Heat | 3 | W | Thermal |

## Avionics/C3

*Table 23232323. Avionics/C3 quantitative I/Os.*

| Origin Subsystem | Input | Amount | Unit | Recipient Subsystem |
|---|---|---|---|---|
| Avionics/C3 | Heat | 90 | W | Thermal |
| Avionics/C3 | Heat | 6 | W | Thermal |
| Avionics/C3 | Heat | 8 | W | Thermal |

## Power/Thermal

*Table 24242424. Power/Thermal quantitative I/Os.*

| Origin Subsystem | Input | Amount | Unit | Recipient Subsystem |
|---|---|---|---|---|
| Power | Power | 237 | W | ECLSS |
| Power | Power | 12 | W | Prop |

| | | | | |
|---|---|---|---|---|
| Power | Power | 80 | W | Prop |
| Power | Power | (Periodic) 457 | W | Prop |
| Power | Power | 1000 | W | PA |
| Power | Power | 2 | W | PA |
| Power | Power | 108 | W | CA |
| Power | Power | 308 | W | EVA |
| Power | Power | 90 | W | Avionics/C3 |
| Power | Power | 6 | W | Avionics/C3 |
| Power | Power | 6 | W | Avionics/C3 |
| Power | Power | 8 | W | Avionics/C3 |
| Power | Power | 165 | W | Thermal |
| Power | Heat | 7 | kW | Thermal |
| Thermal | Heat | 12 | kW | Jettison |

## Structures

*Table 25252525. Structures quantitative I/Os*

| Origin Subsystem | Input | Amount | Unit | Recipient Subsystem |
|---|---|---|---|---|
| Structures | Heat | 4.6 | kW | Thermal |

## GNC/Prop

The propulsion subsystem does not have any quantifiable inputs into the greater system diagram, as shown on its functional block diagram. While the moment gyros provide telemetry data to the C3 subsystem, there is no quantifiable amount of data sent. Similarly, while the propulsion system provides thrust and torque periodically, this is discussed in the onboard propulsion summary below. The propulsion system requires electricity from the power system as shown above, with periodic spikes in power draw during maneuvers.

## Integrated Subsystem FBD

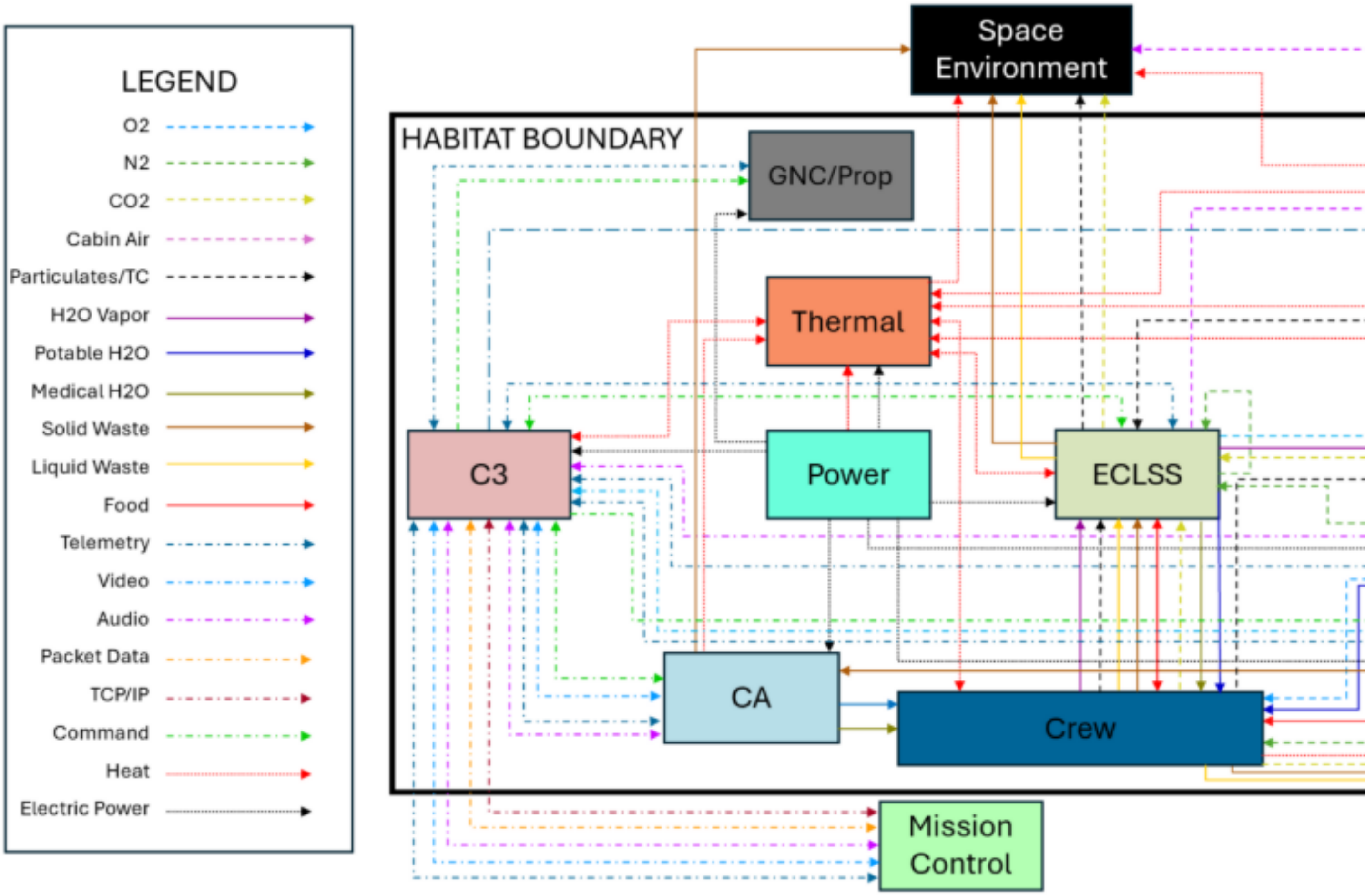


*Figure 13131313. Integrated subsystem interface qualitative I/Os FBD.*

## Logistics Resupply Summary

*Table 26262626. Boyd Station logistics resupply.*

| Equipment ID | Equipment Name | Subsystem | Quantity | Mass Per Unit (kg) | Total Mass (kg) |
|---|---|---|---|---|---|
| ECLSS-01.1 | Pressurized $O_2$ Tank | ECLSS | 1 | 42 | 42 |
| ECLSS-01.2 | $O_2$ | ECLSS | 1 | 29 | 29 |
| ECLSS-02.1 | Pressurized $N_2$ Tank | ECLSS | 1 | 19 | 19 |
| ECLSS-02.2 | $N_2$ | ECLSS | 1 | 10 | 10 |
| ECLSS-05 | LiOH Canister | ECLSS | 21 | 4 | 76 |
| ECLSS-06/07/08/09 | HEPA Filter | ECLSS | 3 | 2 | 6 |

| | | | | | |
|---|---|---|---|---|---|
| ECLSS-07 | Catalytic Filter | ECLSS | 2 | 5 | 11 |
| ECLSS-07 | Activated Charcoal | ECLSS | 2 | 37 | 74 |
| ECLSS-20.1 | Potable Water | ECLSS | 1 | 78 | 78 |
| ECLSS-20.2 | Water Tank | ECLSS | 1 | 16 | 16 |
| ECLSS-21.2 | Medical Water | ECLSS | 1 | 26 | 26 |
| ECLSS-21.1 | Medical Water Tank | ECLSS | 1 | 5 | 5 |
| ECLSS-23.2 | Fecal Canisters | ECLSS | 3 | 2 | 7 |
| ECLSS-24.1, CA-09.1 | Pre-Packaged Food | ECLSS | 21 | 2 | 41 |
| ECLSS-24.2, CA-09.2 | Food Packaging | ECLSS | 21 | < 1 | 9 |
| ECLSS-24.3, CA-09.3 | Margin for Food | ECLSS | 6 | 2 | 12 |
| ECLSS-24.4, CA-09.4 | Packaging for Margin | ECLSS | 6 | < 1 | 3 |
| EVA-02 | LiOH Canister (EVA Specific) | EVA | 3 | 14 | 41 |
| PA-02 | Middeck Locker Payloads | PA | 4 | 25 | 100 |
| CA-01.1 | Personal Hygiene Kit | CA | 21 | 2 | 38 |
| CA-01.2 | Hygiene Supplies (Consumables) | CA | 3 | 1 | 2 |
| CA-04.1 | Kitchen Cleaning Supplies/Trash | CA | 7 | < 1 | 2 |
| CA-04.2 | Disposable Wipes | CA | 21 | < 1 | 6 |
| CA-05 | Medical Aid Kit | CA | 1 | 7 | 7 |
| CA-08.2 | Trash Bags | CA | 42 | < 1 | 2 |
| CA-11 | Personal Preference Kit | CA | 3 | 1 | 2 |
| CA-12.2 | Contingency Human Waste Bags | CA | 21 | < 1 | 5 |
| CA-13.1 | Clothes | CA | 21 | 2 | 48 |
| | | | | Total | 715 |

*Table 27272727. Boyd Station logistic resupply subsystem totals.*

| Subsystem | Total Mass (kg) |
|---|---|
| ECLSS | 462 |
| CA | 112 |
| PA | 100 |
| EVA | 41 |

| | |
|---|---|
| Total | 715 |

## Baseline Habitat Configuration

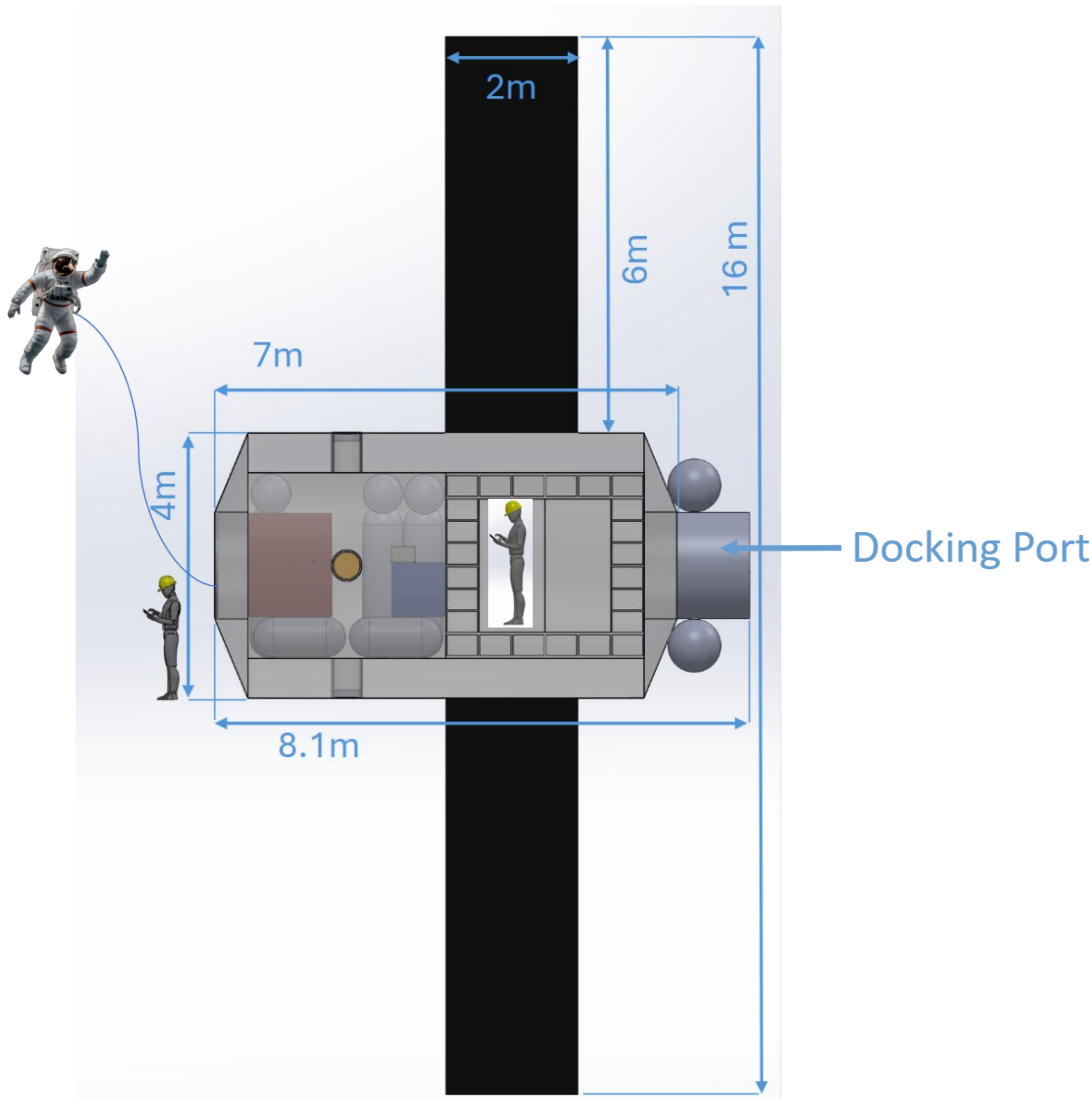


*Figure 14141414: Dimensioned side view showing EVA. For storage considerations the EVA Suits will be stored in the crew transport vehicle pre and post-EVA.*

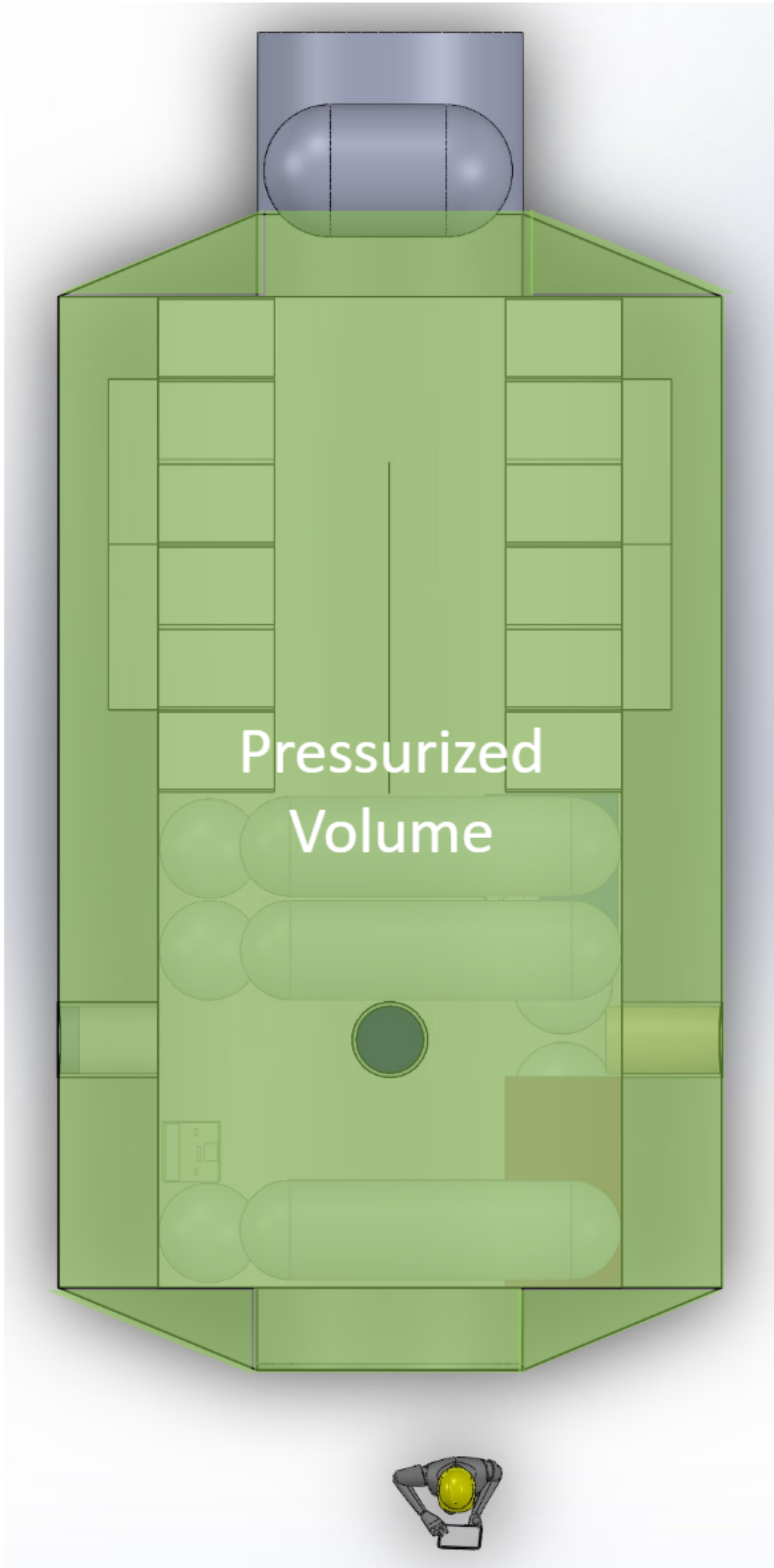


*Figure 15151515: Pressurized volume overlay. Pressurized volume ~82m$^3$*

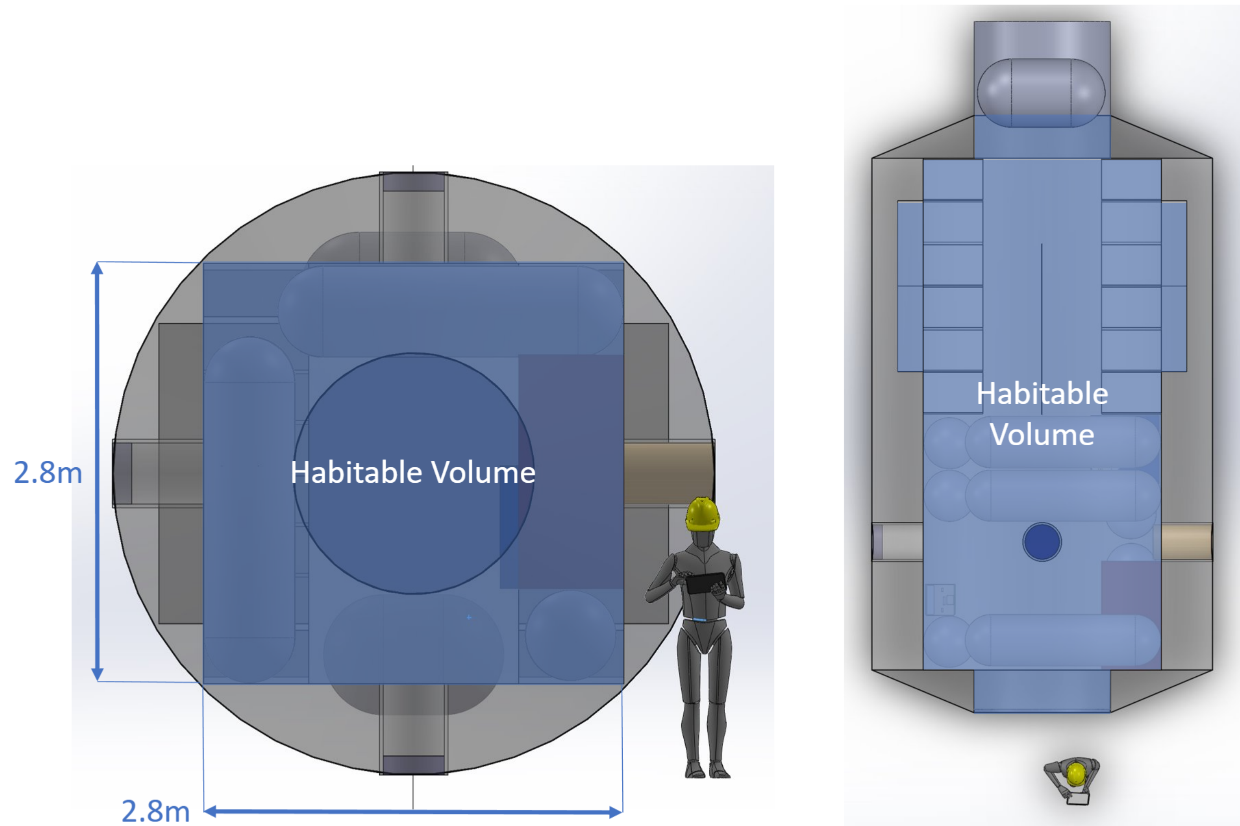


*Figure 16161616: Habitable volume overlay. Habitable volume ~39m³*

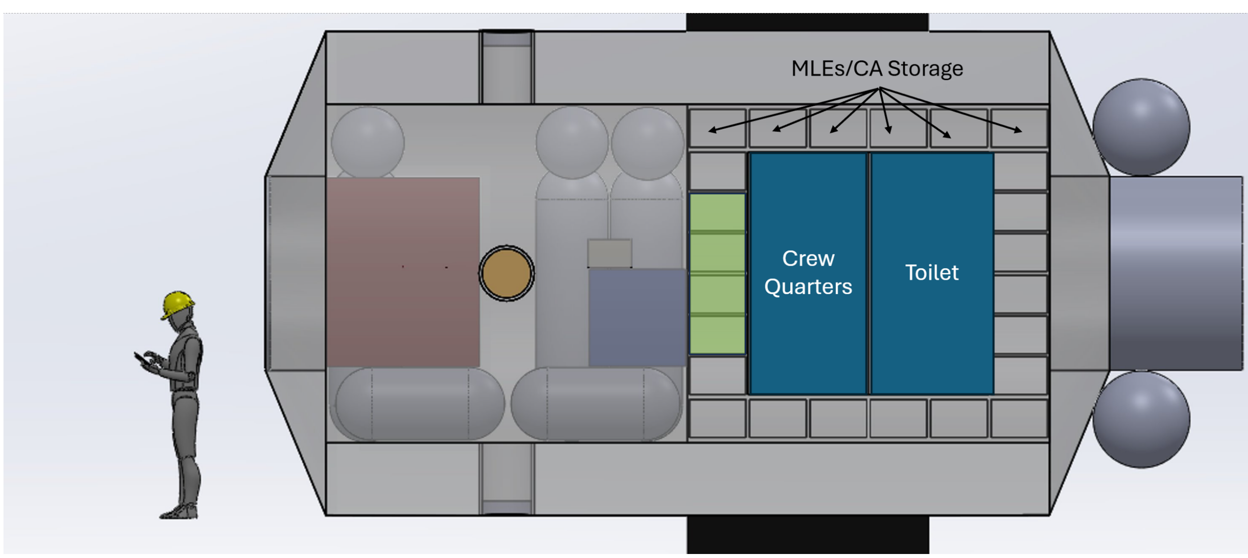


*Figure 17171717: Layout side view. Only four of the MLE storage locations are powered. The remaining 44 "cubbies" are purely for CA storage purposes.*

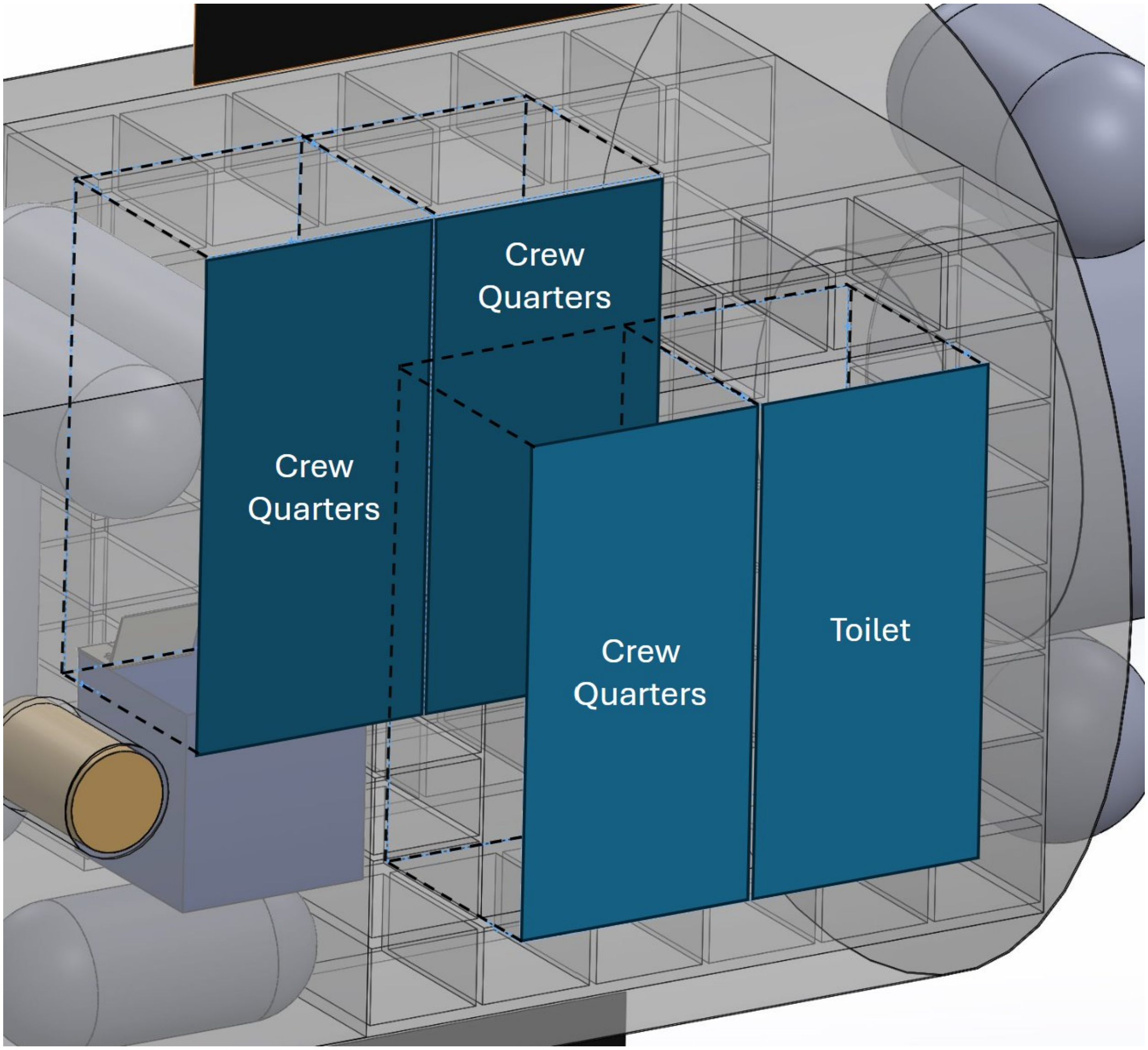


*Figure 18181818: Layout - crew quarters and toilet*

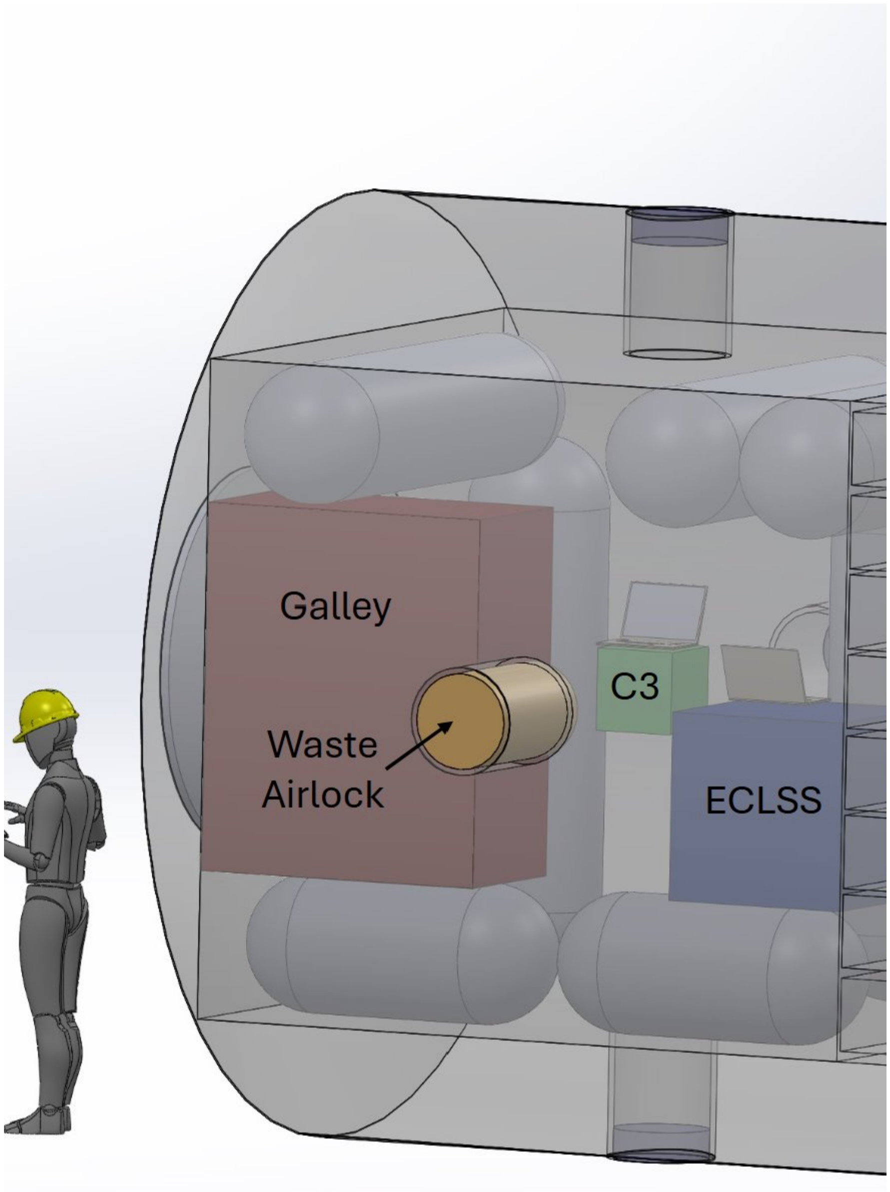


*Figure 19191919: Layout – subsystems*

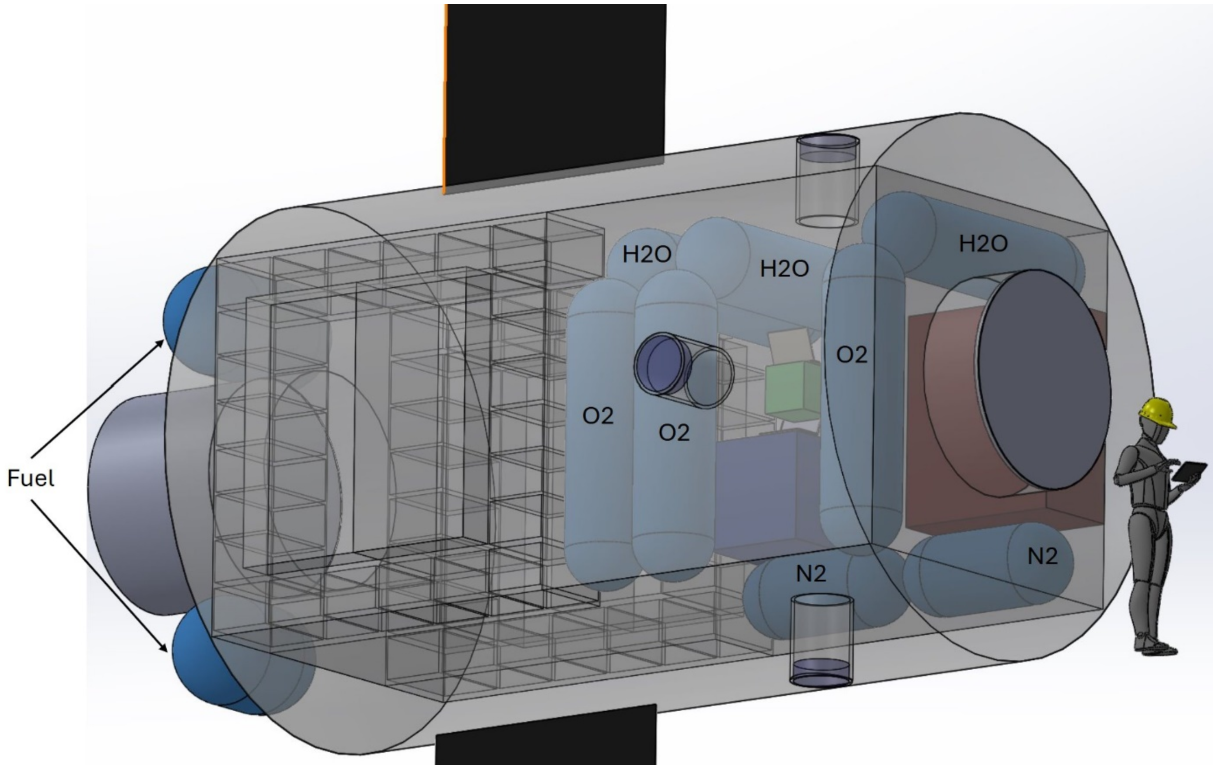


*Figure 20202020: Layout - tanks*

# Conceptual Design Validation

## Summary of Mass Estimates

The top-down mass estimate is based on some of the base factors of the mission put into a formula based on empirical models. The pressurized volume of the station is based on the standard of 5 $m^3$ of lab space per person of lab space, 10 $m^3$ of habitat space per person and a habitable volume to pressurized volume ratio of 67%. These values were then used to calculate the mass of propellant needed on board using the Isp of the propulsion system, 240 seconds. These values are added to give a first-pass burnout mass estimate.

$$Pressurized\ Volume = 3\ Crew \times (5m^3 Hab\ Space + 10m^3 of\ Lab\ Space) \times 1.67 = 75m^3$$

$$m_{bodry} = 592 \times (Number\ of\ Crew \times Duration\ of\ Mission \times Pressurized\ Volume)^{0.346}$$

$$592 \times (3\ Crew\ Members \times 7\ Days \times 75\ m^3)^{0.346} = 7561\ \ kg$$

$$m_{prop} = m_{bodry}\left(\exp\left(\frac{\Delta v}{Isp \times 9.81}\right) - 1\right) = 7561\left(\exp\left(\frac{480}{239 \times 9.81}\right) - 1\right) = 1709\ kg$$

Note: This propellant mass calculation is lower than what is calculated in the propellant summary as the later analysis uses the iterated higher mass of the station and is properly separated between the reboost and attitude thrusters which have different specific impulses.

The process calls to iterate the design by adding the propellant mass to the dry burnout mass and calculate the propellant mass to further refine the mass estimate to include the mass of the propellant on board, giving a more accurate burnout mass estimate. This estimate represents the top-down mass estimate of the station.

$$m_{prop2} = m_{bodry} + M_{prop}\left(\exp\left(\frac{\Delta v}{Isp \times 9.81}\right) - 1\right)$$

$$9527\left(\exp\left(\frac{480}{239 \times 9.81}\right) - 1\right) = 2510\ kg$$

$$m_{bowet} = m_{bodry} + m_{prop2} = 9527kg + 2510kg = 12037kg$$

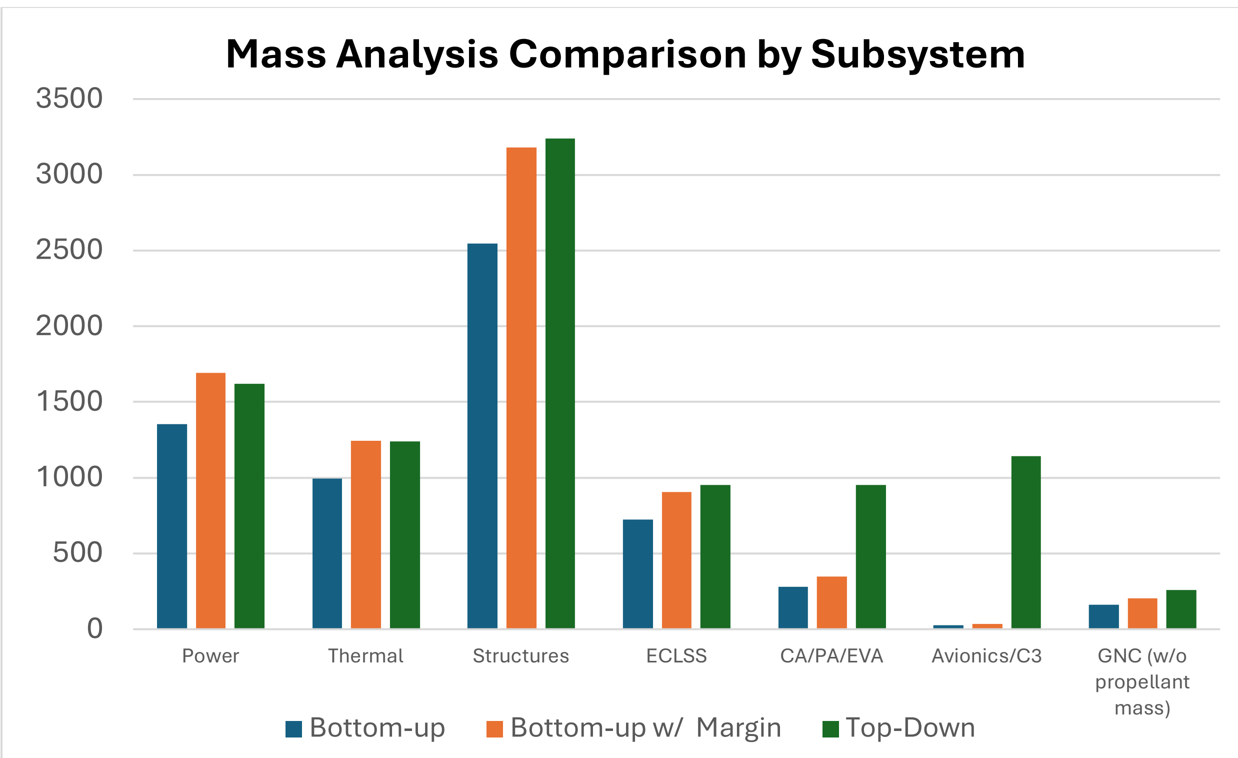


*Figure 21212121. Mass Comparison by Subsystem from bottom-up and top-down approaches.*

*Table 28282828. Boyd Station mass analysis and comparison values rounded to nearest kg.*

| Subsystem | Bottom-up | Bottom-up with Margin | Top-Down |
|---|---|---|---|
| Power | 1353 | 1691 | 1619 |
| Thermal | 995 | 1247 | 1238 |
| Structures | 2546 | 3182 | 3239 |
| ECLSS | 725 | 907 | 952 |
| CA/PA/EVA | 280 | 350 | 952 |
| Avionics/C3 | 29 | 36 | 1143 |
| GNC (w/o propellant mass) | 163 | 204 | 257 |
| Total Dry Mass | 6088 | 7610 | 9403 |

The top-down calculations for each subsystem were based on the Mass Budget provided in Table 12-7 of Human Spaceflight (Larson, 2014) As Boyd Station does not have landing gear, the 18% allocated to this system has been redistributed to all subsystems except for GNC. Each subsystem top-down mass estimate budget has been increased by 2%. The thermal system has received an additional 2%, for a 4% total increase in its top-down mass estimate.

*Table 29292929. Heuristic Top-down mass budget distribution.*

| **Element** | **% Value** |
|---|---|
| Structures | 24 |
| Mechanisms | 10 |
| Thermal Protection | 13 |
| Propulsion System | 15 |
| Attitude Control | 4 |
| Power | 17 |
| Avionics | 12 |
| ECLSS | 10 |
| Crew Accommodations | 10 |
| Total (w/o Propulsion System) | 100 |

Many of the mass discrepancies between the different subsystems are small, usually within a single digit percentage of error. With many of these, such as the structure, power, thermal and ECLSS systems, there are many elements in the design that are too granular to include in a bottom up approach, such as cabling, ducting, plumbing, small structural elements and other small components that are difficult to account for.

The subsystems with the greatest discrepancies are the Crew/Payload accommodations, Avionics and GNC/Prop. Crew accommodations on Boyd station are very slim as the mission duration is very low: only a week. As a result items like exercise equipment have not been included that would be needed to accommodate longer stays. Avionics has a much smaller mass estimate due to a large amount of sensors not being accounted for or parted out to other subsystems. Humidity sensors, pressure monitors and others are distributed to other subsystems and other small components of the avionics subsystem are difficult to account for. Similarly, fuel monitoring and environmental regulation hardware was difficult to account for in designing the propulsion system, which lead to a difference in mass estimates.

## Volumetric Efficiency

The Volumetric Efficiency of Boyd Station can be determined with the following equation:

$$Volumetric\ Efficiency = \frac{Net\ Habitable\ Volume}{Total\ Pressurized\ Volume} \cdot 100,$$

where Net Habitable Volume = 39.150 m$^3$ and Total Pressurized Volume = 81.932 m$^3$ have been extracted directly from the CAD model of Boyd Station. Consequently, we have that:

$$\boxed{\boldsymbol{Volumetric\ Efficiency\ = 47.78\%}.}$$

## Packaging Density

The Packaging Density of Boyd Station can be determined with the following equation:

$$Packaging\ Density = \frac{Total\ Habitat\ Mass\ (with\ margin)}{Total\ Pressurized\ Volume},$$

where Total Pressurized Volume = 81.932 m$^3$ and Total Habitat Mass = 12037 kg is the bottom-up wet mass with 25% margin. Note that the mass of the crew is not included in this calculation.

$$\boxed{\boldsymbol{Packaging\ Density = \ 147\ \mathrm{kg/m^3}}}$$

## Consumable Mass Fraction

The Consumable Mass Fraction of Boyd Station can be determined with the following equation:

$$Consumable\ Mass\ Fraction = \frac{Total\ Dry\ Mass}{Total\ Wet\ Mass\ (with\ margin)} \cdot 100,$$

where Total Wet Mass = 12037 kg is the bottom-up wet mass with 25% margin, and the Total Dry Mass is the difference between the wet mass of the habitat and the mass of the consumables. First, we calculate the total mass of the consumables by adding the mass of the consumables from each subsystem:

$$Consumables = C_{ECLSS} + C_{CA} + C_{PA} + C_{EVA} + Fuel = 2893 \text{ kg},$$

where $C_{ECLSS}$ = 462 kg, $C_{CA}$ = 112 kg, $C_{PA}$ = 100 kg, and $C_{EVA}$ = 41 kg are found in Table 27272727Table 2727, and $Fuel$ = 2178 kg is detailed in Table 30303030Table 3030. Next, the Total Dry Mass of Boyd Station can be calculated:

$$Total\ Dry\ Mass = Total\ Wet\ Mass\ (with\ margin) - Consumables = 9145 \text{ kg}.$$

With a total dry mass of 9145 kg and a total wet mass of 12037 kg, Boyd Station has the following Consumable Mass Fraction:

$$\boxed{\boldsymbol{Consumable\ Mass\ Fraction = 75.97\%}}$$

## Onboard Propulsion Summary

### Delta-V and Prop Budget

Orbital decay modeling was based on an atmospheric model varying with time close to solar minimum. The drag characteristics used were the stations side on profile with solar panels deployed and a drag coefficient of 2.1, similar to Mir's drag coefficient. The stations maximum burnout mass from top-down calculations was used to determine a lower bound delta-v (10100 kg), and the upper bound determined by the mass of the station based on the Master Equipment List (8300 kg) Using the MR-104's designed Isp of 239 seconds, the propellant mass for these maneuvers was also determined, which is ~80 kg for the bottom-up station mass and ~70 kg for the top-down station mass.

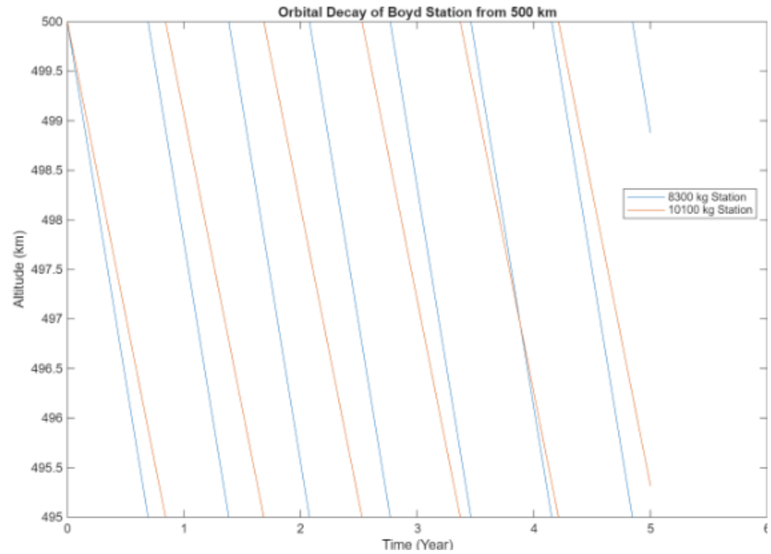


*Figure 22222222. Station Orbital Decay Cases.*

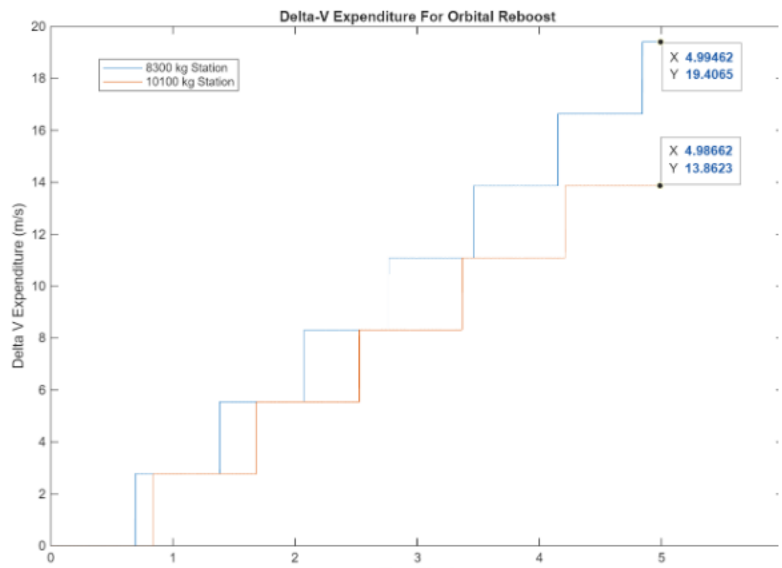


*Figure 23232323. Reboost Delta-V Expenditure.*

## MMOD Avoidance Budget

NASA's MMOD Protection design handbook includes several useful metrics for fuel budgeting for MMOD avoidance. While smaller debris can be hard to track, anticipate, and avoid, larger pieces are more visible and more dangerous to the station. From the beginning of the ISS' operation to June 2023, the station performed 40 propulsive maneuvers to avoid debris (Christiansen, 2009). These maneuvers cost in the range of 0.5-1 m/s, as moving the orbit of the station to avoid debris only involves a small correction. As the handbook cites the debris flux being 50% higher in a 98-degree orbit compared to the

ISS' 51.6 degrees, increasing the amount of yearly avoidance maneuvers accordingly is a reasonable place to start with an avoidance budget. It is also worth building in a growth factor of 25% more delta-v as orbital debris flux increases over the 5-year period. With a count of 2 maneuvers per year, adding a third maneuver to account for the 50% extra flux, and a 25% growth factor, the total annual delta-v needed is ~5 m/s.

## Demise Analysis and Budget

The station will need to perform an altitude lowering maneuver from 500 km down to a 250 km parking orbit. The parking orbit will be achieved to ensure the station will reenter at a planned moment. The final maneuver required for entry will be lowering the periapsis into Earth's atmosphere, at about 90 km. These three maneuvers will cost about 240 m/s of delta-v total, and will use about 850 kg of propellant from the fuel system. Additionally, 15 m/s of maneuvering fuel will be included to ensure the station can maintain attitude during deorbit maneuvers.

## Station Keeping Budget

Boyd station will primarily be using a set of Control Moment Gyros (GMGs) for attitude control. A set of four Blue Canyon CMG-12s will be able to provide a peak torque of 48 Nms on the station, and redundant control with the gyros being able to provide torque in all three axes. While these will be able to provide ample station keeping, reaction will be necessary to desaturate the gyros occasionally and allow for increased rotation authority during station phases, such as docking procedures or reboost maneuvers. The ISS has propellant use figures from the first decade of its operation, broken down by usage. Based on the propellant usage figures following the installation of the Z1 Truss segment with the station's CMGs, the ISS uses on average 13.4% of its propellant budget every year for attitude adjustment, with most of the rest used for reboost maneuvers (Johnson, 1998) This amounts to 25 m/s per year for 125 m/s total.

## Prop System Characteristics

Boyd Station will use a monopropellant system driven by mono-methyl hydrazine (MMH) for both its attitude keeping and reboost engines. The propellant will be able to be stored together in tanks near the docking adapter. Storing tanks on the exterior of the station keeps the crew separate from the toxic propellant both inside the habitat and during EVAs which occur at the other end of the station.

The reboost engines are two Rocketdyne MR-104s mounted to provide prograde thrust (Hsu, 2023). Together they can deliver between 0.5-1 kN of thrust to counteract drag and solar radiation pressure and keep the orbital altitude of 500 km. With a specific impulse of 239 seconds, they allow for a high efficiency of a monopropellant system. Eight Rocketdyne MRM-122 clusters provide attitude control for the station (Wilson, 2020). These are also monopropellant thrusters, each providing up to 142 N and 228 seconds of isp. These are positioned orthogonally in a ring around the fore and aft of the station for attitude control.

*Table 30303030. Summary Of Delta V and Propellant Masses.*

| **Impulse Need** | **Delta-V (m/s)** | **MD-104 Prop Mass (kg)** | **MRM-122 Prop Mass (kg)** | **Total Prop Mass (kg)** |
|---|---|---|---|---|
| Deorbit Fuel | 240 | 1088 | 0 | 1089 |
| Station Keeping | 125 | 0 | 581 | 5801 |
| Demise Attitude | 15 | 0 | 68 | 68 |
| Altitude Keeping | 25 | 108 | 0 | 108 |
| MMOD Avoidance | 25 | 54 | 57 | 111 |
| Growth Margin | 50 | 108 | 113 | 221 |
| **Total** | 480 | 1359 | 819 | 2178 |

## Launch Vehicle Selection

A Sun-Synchronous Orbit (SSO) for a 500-kilometer orbit sits at approximately 97 degrees, which is best achieved from launch sites capable of supporting polar launches. While there are a few future launch vehicle options out of Vandenberg, these are not yet operational and, in some cases, do not have available metrics for launch mass to orbit. Vehicles like Vulcan Centaur and New Glenn have higher performance than required for Boyd station. Additionally, these vehicles are still new in their launch campaigns and might incur high cost while still having relatively low reliability.

SpaceX's Falcon 9 represents the best choice based on these design drivers. It has a long history of flying out of Vandenberg, a high launch cadence from that site, a low launch price and more than enough capability to perform the mission. While SpaceX does not advertise an upper bound of Mass-To-SSO from Vandenberg, there is a history of Starlink launches from 2025's Group 17 that have delivered 17.5 tons to a 560 km SSO.

While our burnout mass in top-down and bottom-up both sit far beneath this seventeen-ton threshold, Falcon 9 is more than capable.

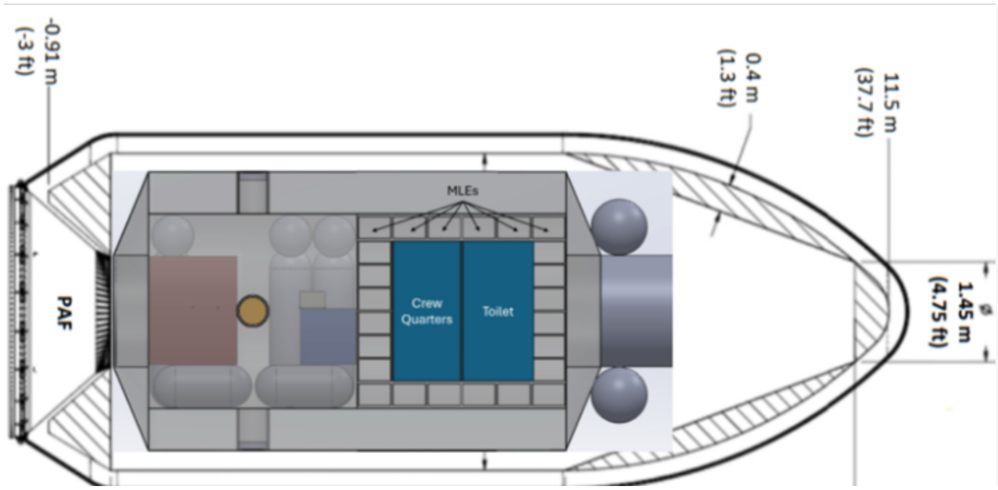


*Figure 24242424. Boyd Station Encapsulated in Falcon 9's Standard Fairing.*

Falcon 9's reusability is a useful feature: the Starlink launches that the mass budget is based upon include the first stage booster landing and being reused. While the station as designed is nowhere near the seventeen-ton limit, the booster could be expanded to ensure a slightly larger burnout mass if the mass of the station has crept beyond initial estimates. While finance is not a large design driver for the project brief, the fact that a Falcon 9 launch sits about an order of magnitude lower in cost than competitors make it an attractive option.

# Conclusion

## Key Design Features and Risk Considerations

Our group's decision to place the station into a Sun-synchronous orbit has had cascading consequences for the rest of our design process. Firstly, launching into a polar orbit on American soil is most easily done from Vandenberg AFB. As SpaceX's Falcon 9 is the most capable launch vehicle with launch cadence from Vandenberg, it is the obvious choice of launch vehicle in addition to its convenient orbital capabilities. An SSO also has increased delta-V requirements, but the station mass is well within the capabilities of the Falcon 9. The countervailing advantage of such an orbit is drastically simplified power and thermal requirements. Rather than sizing solar panels, batteries, and radiators for alternating periods of darkness and full sunlight exposure, the station will be continuously bathed in twilight, relegating batteries to a purely emergency role and simplifying thermal flux to steady-state energy transfer. A theoretical advantage of such an orbit is reduced stress on the station due to the elimination of thermal expansion and contraction cycles, but these benefits are not explored in this analysis.

Additionally, this station is not equipped with an airlock and instead is provisioned with extra inert gas and oxygen to depressurize and repressurize the station for the purpose of performing EVAs. Compared to the mass of an airlock, the mass of the nitrogen on board the station is surprisingly small, and results in an increased budget for other systems. It also increases habitable volume for the same pressurized volume. Further reducing station mass is the decision to operate the station at a reduced total pressure, similar to a continuous EVA camp-out atmosphere, to reduce pre-breathe time for the astronauts. The tradeoff of removing an airlock is the necessity to vacuum-harden all internal systems of the station.

One final feature of the station atmosphere is the implementation of a mono-directional flow atmosphere, where in the absence of convection currents, onboard fans provide continuous flow towards the back of the station, before the air is circulated via vents to ECLSS systems and then reintroduced to the environment. Not only does this system effectively circulate air throughout the station, but by placing crew quarters closest to ECLSS exit vents, astronauts will be guaranteed the best quality and most conditioned air available. In case of a fire or other atmospheric emergency onboard the station during rest hours, crew will have the minimum chance of breathing contaminated air before they are awakened by alarms and are able to control the emergency, hopefully resulting in reduced response time.

There are a few general risk considerations that are worth discussing in regard to the station. The orbit design of the station leaves the crew uniquely vulnerable to radiation during high energy solar events like Solar Particle Events. In these scenarios, it would be ideal to have the crew perform an abort the mission, as the scope of the mission is quite short. There is some logistical risk associated with the design decision to not include an airlock and instead evacuate the cabin. This decision was made to reduce the mass of the station and there is a large amount of oxygen and nitrogen aboard to compensate for the loss of cabin atmosphere, but there is risk is materials like food and personal belongings being damaged during cabin evacuation, which is why time has been built into the EVA schedule to secure the cabin before the EVA. While there is danger associated with only having a hatch that opens into vacuum and not an airlock, pressure monitors will be able to warn the crew of a leak before they are in danger.

Finally, there are a number of failures on board the station that, while unlikely, could be catastrophic to station and crew health like power loss and avionics failure. As discussed below, one of the recommendations for future work is an analysis of failure modes for these critical systems and how redundancy for critical systems along with a high degree of fault tolerance can be built into the station. This work would decrease the already low likelihood of these events and even decrease the consequence of these events as well.

## Business Plan

Boyd station is a commercial station that will serve as a scientific research station in the space environment, with special emphasis on radiation research due to its unique polar orbit. Boyd station will offer multiple points of consumer interaction to conduct scientific research. As the station incorporates multiple sensors and cameras that will continue operations during crewed and uncrewed periods, Boyd station will have a significant dataset of environmental readings and images. These images and data will be available for use in research studies on a subscription-based platform. For experiments specifically tailored for the space environment, Boyd station will operate similarly to how the ISS currently handles its scientific payloads. With space for a minimum of four MLEs, organizations can purchase a spot for their experiment for the duration of one of the ten missions set to occur during the five-year lifespan of Boyd Station. For companies or organizations that would like more hands-on experience in the space environment, they can purchase berths for one of the ten missions for one, two, or all three crew members. The margin for small scale delta-V maneuvers has been planned prior to the launch of Boyd Station. Large scale maneuvers like orbital maintenance and reboost procedures will be conducted when the crew vehicle is disconnected from the station. While keeping the crew vehicle attached to Boyd station during each mission will cost more than if it were detached, the benefits regarding risk mitigation in emergency events such as an abort procedure outweigh the financial costs.

## Key Take Aways

The first major insight gained from this design project was how easily scope creep can influence and complicate a system's development. During the early design sessions, the team proposed additional capabilities, such as continuous biological testing, to enhance the station's functionality. While these ideas added appeal, they also introduced unnecessary mass, power consumption, and design complexity that fell outside the project's actual requirements.

Another key insight was the importance of maintaining a focused design. Establishing clear priorities helped prevent feature creep from derailing the project. With a well-defined scope, the team could evaluate new proposals and determine whether they were truly essential or merely desirable additions that would burden the design.

The final insight was the realization that the support systems designed to meet the project's objectives were far less demanding than initially expected. Because the station only needed to support a crew of three for short, infrequent one-week periods, the complexity, mass, and power needs of life-support and accommodation systems were significantly reduced. This understanding helped streamline the overall design and avoid overengineering.

## Recommendations for Future Steps

For a continuation of this project, picking up where we left off, we would recommend refining the interior volumetric sketches to include a finer detailed visual representation of the habitat and all its components. Current mass, power and volume calculations do not include consideration of small system connections such as wiring or ECLSS system tubing. To rectify this our next step would be to confirm the necessary connections and calculate their full mass, power, and volume specifications. We would also recommend further refinement of the current technology specifications by contacting suppliers for exact value. Our last recommendation is to begin a detailed failure mode analysis with a high-level fault tree.

# Acknowledgements

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